# Thermodynamics – Division C

Invitational Reference Binder (unofficial study reference)

Built 2026-10-21 for a tournament on **Thursday, 22 October 2026** ("tomorrow" resolved in America/Chicago; adjust if your time zone differs)

Team / school: _____ Tournament: _____

Partner A: _____ Partner B: _____

Suggested split (adaptable): A — Tabs 3–4 (gases, laws, cycles); B — Tabs 2, 5, 7 (heat, transfer, data); share Tabs 1 and 6.

**RULE-DEPENDENT RULES STATUS — READ BEFORE THE EVENT.** The official 2027 Thermodynamics rules, official clarifications, and this tournament's instructions were **not inspected**: none were attached, and the official site was blocked from the build environment. Therefore the following are **UNVERIFIED**:

1. whether this binder may be used at all, and in what form (page or sheet limit, binding, protectors, tabs, two-sided printing);
2. which calculator class is allowed;
3. the written-test scope or topic list (topic labels below are editorial, not official);
4. device construction, procedure, timing, and scoring (p. 32);
5. any modifications announced by the tournament host.

Snippet-level information only (not verified): the official event description reportedly reads "construct a device prior to the tournament designed to *collect heat* and complete a written test on thermodynamic concepts,"[SO1] while a community wiki describes a *heat-retention* beaker task.[WIKI] The conflict is unresolved. **An unverified permission is not a scientific error:** the physics in this binder was checked independently (p. 39); its *eligibility* for use was not.

**Table 0.1.** Alignment summary (full matrix in `rules-alignment.md`).

| Item | Status | Location |
|---|---|---|
| Written-test scope | unverified | whole binder; editorial labels |
| Reference / calculator permissions | unverified | this page |
| Device participation and limits | unverified; sources conflict | p. 32 |
| Event in 2027 Div. C slate | snippet only | p. 39 |

### Pre-event checklist (most important first)

- ☐ Confirm in the 2027 Rules Manual (and host instructions) that a binder is allowed, and its size and format limits. Remove pages if required.
- ☐ Confirm the calculator class.
- ☐ Confirm device specs, impound time, energy source, and what is scored before touching the device (p. 32).
- ☐ Check for official clarifications/FAQs and host modifications.

### Content labels (editorial, not official):

**[CORE]** prioritized foundation; **[EXTENSION]** useful added depth, no claim it is tested; **[RULE-DEPENDENT]** construction, scoring, permissions, or host-specific.

### Box types:

**FORMULA** canonical equations with IDs.

**KEY IDEA / DECISION** model choice and decision steps.

**SOLVED DEMONSTRATION** short, fully solved, numerically checked.

**TRAP / LIMIT** common mistakes and validity limits.

**Source tags** (superscript, clickable): e.g. [IAPWS], [CODATA], [DER] (derived/checked here), [EK] (editor knowledge, unverified this build). Register: p. 39.

**Conventions:** $\Delta U = Q - W_{\text{by}}$ (p. 16); kelvin in ratios and powers; pressures absolute; cycle quantities are magnitudes.

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**How to use:** (1) Find the cue in the **question router** (p. 3) or the **index** (p. 40). (2) For a calculation, use the **formula selector** (p. 4) to confirm the equation's validity and hazards, then jump to its canonical page. (3) Data tables: Tab. 7.1, p. 34. Equation and table numbers start with the tab number (Eq. (3.20) is in Tab 3).

# TAB 1 Quick Reference

## 1.1 Question router

Find the physical cue (synonyms included), then jump. Entries give the canonical equation number with its page, or a table/section page. All links are clickable in the PDF.

### Heat and calorimetry

**Mix / equilibrium (final) temperature / method of mixtures** → ledger (2.12) p.9, shortcut (2.13) p.9

**Ice added / partial melt / ice left over / freezes** → decision p. 10; water data Tab. 2.5

**Steam condenses / steam burn / steam into water** → p. 10; $h_{fg}$ Tab. 2.7

**Unknown metal / specific heat / calorimeter constant / water equivalent** → demos p. 10; Tab. 7.2; Dulong–Petit p. 9

**Heater / electrical energy / joule heating / $VIt$** → (2.16) p.10

**Coffee cup / bomb calorimeter / solution or combustion energy / $\Delta U$ to $\Delta H$** → (2.14) p.10, (2.15) p.10, (6.4) p.29

**Heating curve / plateau / latent vs sensible** → Fig. 2.1, p. 10

### Temperature, expansion, thermometers

**Convert °C, °F, K, R / absolute zero** → (2.1) p.7–(2.4)

**Thermometer type / thermocouple / RTD / thermistor / IR gun / calibration** → Tab. 2.2 p. 8; (2.5) p.8

**Expansion / rail gap / ring on shaft / hole / overflow / thermal stress / bimetal** → (2.6) p.8, (2.8) p.8, (2.9) p.8

**Water densest at 4 °C / lakes freeze on top / ice floats** → p. 8

### Phases

**Phase diagram / triple point / critical point / supercritical / dry ice** → Fig. 2.3 p. 11

**Boiling point at altitude / pressure cooker / vapor pressure vs $T$** → Tab. 2.8 p. 13; (2.20) p.12

**Melting under pressure / skating / slope of a phase line** → (2.19) p.12

**Humidity / dew point / frost / fog / condensation on a glass** → (2.21) p.12; Tab. 2.9

**Gas collected over water / wet gas** → (2.23) p.12

**Quality / wet steam / two-phase / lever rule** → (2.18) p.12

**Evaporative cooling / sweating / open beaker / supercooling** → p. 12

### Gases

**Ideal gas / density / molar mass from density / absolute pressure** → (3.1) p.14, (3.2) p.14

**STP / SATP / molar volume 22.4 or 22.7** → Tab. 3.1 p. 14

**Partial pressure / mole fraction / Dalton / air composition** → (3.4) p.14

**rms / mean / most probable speed / Maxwell distribution** → (3.7) p.14, Fig. 3.1

**Effusion / Graham / pinhole leak / diffusion** → (3.10) p.14

**Mean free path / collision frequency** → (3.11) p.14

**$C_v$, $C_p$, $\gamma$ / degrees of freedom / equipartition** → Tab. 3.2, (3.12) p.15, Tab. 3.3

**Real gas / van der Waals / compressibility $Z$ / critical constants** → (3.14) p.15, Tab. 3.4

### First law

**Constant $P$ / rigid tank / isothermal / insulated piston / adiabatic** → master Tab. 1.2 p. 5

**PV area / work by gas / piston / sudden expansion / vacuum** → (3.19) p.16–(3.21)

**Sign of $W$ / chemistry $w$ / polytropic $PV^n$** → Tab. 3.5 p. 16; (3.27) p.16

**Turbine / compressor / pump / nozzle / mixing / heat exchanger** → Tab. 3.6, (3.29) p.17

**Throttling / Joule–Thomson / valve / inversion / free expansion** → (3.31) p.17, Tab. 3.7

### Entropy, free energy, equilibrium

**Entropy change / irreversible / universe / generation** → (3.34) p.18; collection (3.35) p.18

**Spontaneous / $\Delta G = \Delta H - T\Delta S$ / crossover temperature** → (3.42) p.19, Tab. 3.8

**Equilibrium constant / $Q$ vs $K$ / $\Delta G^\circ = -RT \ln K$** → (3.47) p.19; van 't Hoff (6.6) p.29

**Third law / absolute zero / residual entropy / microstates** → p. 19; (3.40) p.19

### Cycles

**Efficiency / COP / Carnot / "possible machine?"** → (4.3) p.20; algorithm p. 20

**Otto / Diesel / Brayton / compression, cutoff, pressure ratio** → (4.4) p.21–(4.6); Tab. 4.1

**Steam plant / Rankine / refrigerator / A/C / heat pump / R-134a** → (4.7) p.21, (4.8) p.21

**Clockwise loop / multi-leg cycle / net work** → p. 22

### Heat transfer

**Wall / pipe / layers / R-value / U-factor / thermal bridge** → (5.7) p.23, (5.9) p.23, Tab. 5.2

**Airflow / wind / fan / $h$ / film temperature / boundary layer** → (5.13) p.25; Tab. 5.3 p. 26

**Natural convection / buoyancy / Rayleigh / Grashof / beaker** → (5.16) p.25; demo p. 25

**Glow / color / peak wavelength / Wien / sunlight vs IR** → (5.19) p.26

**Emissivity / shiny / blackbody / shield / Dewar / foil** → (5.17) p.26, (5.23) p.27, Tab. 5.4

**Cooling time / Newton cooling / half-life / time constant / fitted slope / Biot** → (5.26) p.27, (5.29) p.27, (5.28) p.27

### Chemistry, history, device, data

**Reaction enthalpy / formation / Hess / combustion / bond energies** → (6.1) p.29; Tab. 6.1; p. 29

**Person / date / experiment / discovery** → Tab. 6.2 p. 31

**Device prediction / calibration / ambient correction** → p. 32; (6.7) p.32

**Constants / $R$ in other units / unit conversion / air or water properties** → Tab. 7.3, Tab. 7.4, Tab. 7.5, Tab. 7.6



## 1.2 Formula selector

Cue → equation → when it is valid → the usual unit or sign hazard → full explanation. Equation numbers are the canonical ones in the chapters; this table repeats only the most-used families.

| Cue | Equation | Valid when | Unit / sign hazard | Eq., page |
|---|---|---|---|---|
| **Energy, calorimetry, phases** | | | | |
| Closed system | $\Delta U = Q - W_{\text{by}}$ | any closed system | chemistry uses $w = W_{\text{on}}$ | (3.17) p.16 |
| Mixing | $\sum_i \Delta H_i = E_{\text{in}} - Q_{\text{loss}}$ | const. $P$, all bodies listed | signed $(T_f - T_i)$; include cup | (2.12) p.9 |
| Sensible heat | $Q = mc\Delta T$ (or $\int C\,\mathrm{d}T$) | one phase, $c$ roughly const. | g vs kg; J vs cal | (2.10) p.9 |
| Phase change | $Q = \pm mL$ | at $T_{\text{tr}}(P)$ | $L_v$ depends on $T$ | p. 9 |
| Vapor pressure | $\ln \frac{P_2}{P_1} = -\frac{\Delta \bar{h}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$ | ideal vapor, $v_g \gg v_f$, $\Delta h$ const. | kelvin; molar $\Delta \bar{h}$ | (2.20) p.12 |
| Phase-line slope | $\frac{\mathrm{d}P}{\mathrm{d}T} = \frac{\Delta h}{T \Delta v}$ | exact on coexistence | sign of $\Delta v$ | (2.19) p.12 |
| Humidity | $\phi = p_v / p_{\text{sat}}(T)$; $p_{\text{sat}}(T_{\text{dp}}) = p_v$ | ideal-gas vapor | dew point ≠ wet bulb | (2.21) p.12 |
| Quality | $y = y_f + x(y_g - y_f)$ | inside dome only | mass fraction | (2.18) p.12 |
| **Gases and processes** | | | | |
| Ideal gas | $PV = nRT = NkT = mR_sT$ | low density ($Z \approx 1$) | absolute $P$, K | (3.1) p.14 |
| Speeds | $v_{\text{rms}} = \sqrt{3RT/M}$ etc. | ideal gas, equilibrium | $M$ in kg/mol | (3.7) p.14 |
| Effusion | rate $\propto 1/\sqrt{M}$ | same $T$, $n_N$, hole | not exact for diffusion | (3.10) p.14 |
| Mean free path | $\lambda = kT/(\sqrt{2}\pi d^2 P)$ | hard spheres | number density | (3.11) p.14 |
| Mayer | $\bar{C}_p - \bar{C}_v = R$ | ideal gas | molar vs total vs specific | (3.12) p.15 |
| Ideal-gas $\Delta U$ | $\Delta U = n\bar{C}_v \Delta T$ | *any* ideal-gas process | not "$Q$" | (3.23) p.16 |
| Work | $W_{\text{by}} = \int P_{\text{ext}} \mathrm{d}V$ | always | use gas $P$ only if quasistatic | (3.19) p.16 |
| Isothermal work | $W = nRT \ln(V_2/V_1)$ | reversible, ideal gas | ln of a ratio | (3.24) p.16 |
| Adiabat | $PV^\gamma = \text{const.}$ | reversible, ideal, const. $\gamma$ | not for free expansion | (3.25) p.16 |
| Polytropic | $W_{\text{by}} = \frac{P_2V_2 - P_1V_1}{1 - n_p}$ | reversible $PV^{n_p}$ | $n_p = 1$: logarithm | (3.27) p.16 |
| Flow device | $\dot{Q} - \dot{W} = \dot{m}\,\Delta(h + \frac{V^2}{2} + gz)$ | steady flow | kJ vs J for $V^2/2$ | (3.29) p.17 |
| Throttling | $h_2 = h_1$; $\mu_{\text{JT}} = (\partial T/\partial P)_h$ | adiabatic valve | ideal gas: $\Delta T = 0$ | (3.31) p.17 |
| van der Waals | $(P + a/\bar{V}^2)(\bar{V} - b) = RT$ | qualitative real gas | units of $a$, $b$ | (3.14) p.15 |
| **Entropy, free energy, cycles** | | | | |
| Entropy balance | $\Delta S = \sum \int \delta Q/T_b + S_{\text{gen}}$ | $S_{\text{gen}} \geq 0$ | $T_b$ in K at boundary | (3.34) p.18 |
| Ideal-gas $\Delta S$ | $n\bar{C}_v \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1}$ | state function | ratios in K | (3.35) p.18 |
| Solid/liquid $\Delta S$ | $mc \ln(T_2/T_1)$ | incompressible | not $mc\Delta T/T$ | (3.36) p.18 |
| Phase $\Delta S$ | $\Delta H_{\text{tr}}/T_{\text{tr}}$ | equilibrium $T_{\text{tr}}$ | supercooled: use path | (3.37) p.18 |
| Gibbs | $\Delta G = \Delta H - T\Delta S$ | constant $T$ | J vs kJ | (3.42) p.19 |
| Equilibrium | $\Delta_r G^\circ = -RT \ln K$ | activities | $K$ dimensionless | (3.47) p.19 |
| Carnot | $\eta_C = 1 - T_L/T_H$ | reservoir $T$ in K | not endpoint averages | (4.3) p.20 |
| COP | $\text{COP}_{HP} = \text{COP}_R + 1$ | same machine | magnitudes | (4.2) p.20 |
| Otto | $\eta = 1 - r^{1-\gamma}$ | cold-air standard | $r = V_{\max}/V_{\min}$ | (4.4) p.21 |
| Brayton | $\eta = 1 - r_p^{-(\gamma-1)/\gamma}$ | cold-air standard | pressure ratio | (4.6) p.21 |

| Cue | Equation | Valid when | Unit / sign hazard | Eq., page |
|---|---|---|---|---|
| Vapor cycles | $\eta = \frac{w_t - w_p}{q_{\text{in}}}$, COP $= \frac{h_1 - h_4}{h_2 - h_1}$ | table properties | reference states differ | (4.7) p.21 |
| **Heat transfer and chemistry** | | | | |
| Conduction | $\dot{Q} = kA\Delta T/L$ | steady, 1D, const. $k$ | flux vs rate | (5.2) p.23 |
| Networks | $\dot{Q} = \Delta T/\sum R$; $R = L/(kA)$, $1/(hA)$ | steady 1D paths | parallel: add $1/R$ | (5.7) p.23 |
| Cylinder wall | $R = \ln(r_2/r_1)/(2\pi kL)$ | radial, steady | $U$ area basis | (5.4) p.23 |
| R-value | $R_{\text{US}} = 5.678\, R_{\text{SI}}$ | per unit area | direction of factor | (5.9) p.23 |
| Convection | $\dot{Q} = hA(T_s - T_\infty)$ | $h$ from correlation | $h$ not a property | (5.13) p.25 |
| Groups | $Re$, $Pr$, $Nu$, $Gr$, $Ra$, $Bi$ | correlation-specific $L$ | fluid vs solid $k$ | (5.15) p.25 |
| Radiation (small body) | $\dot{Q} = \varepsilon\sigma A(T_s^4 - T_{\text{sur}}^4)$ | large surroundings | kelvin | (5.20) p.27 |
| Radiation coefficient | $h_r = \varepsilon\sigma(T_s + T_{\text{sur}})(T_s^2 + T_{\text{sur}}^2)$ | combine with $h$ | $4\varepsilon\sigma T_m^3$ only if close | (5.21) p.27 |
| Two surfaces | $\dot{Q} = \sigma\Delta T^4/[\ldots]$ | gray, diffuse | view factor | (5.23) p.27 |
| Wien | $\lambda_{\max} T = 2898\,\mu\text{m}\cdot\text{K}$ | per-wavelength | not per-frequency | (5.19) p.26 |
| Lumped cooling | $\frac{T - T_\infty}{T_i - T_\infty} = e^{-t/\tau}$, $\tau = \frac{C}{hA}$ | $Bi < 0.1$ | $L_c = V/A_s$ | (5.26) p.27 |
| Constant input | $T = T_a + \frac{P}{K} + (T_0 - T_a - \frac{P}{K})e^{-t/\tau}$ | const. $P, K, C$ | phase limits | (5.30) p.28 |
| Reaction enthalpy | $\sum \nu\Delta H_f^\circ(\text{prod}) - \sum \nu\Delta H_f^\circ(\text{reac})$ | phases labeled | $H_2O$(l) vs (g) | (6.1) p.29 |
| Bomb → enthalpy | $\Delta H = \Delta U + \Delta n_{\text{gas}} RT$ | ideal gases | count gas moles only | (6.4) p.29 |
| van 't Hoff | $\ln \frac{K_2}{K_1} = -\frac{\Delta H^\circ}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$ | $\Delta H^\circ$ const. | kelvin | (6.6) p.29 |

## 1.3 Master process table (ideal gas)

**Table 1.2.** Master process table: $n$ moles of ideal gas, constant $\bar{C}_v$, $\bar{C}_p = \bar{C}_v + R$, $\gamma = \bar{C}_p/\bar{C}_v$; work *by* the gas; reversible unless stated. Every row is consistent with $\Delta U = Q - W_{\text{by}}$.[DER]

| Process | $W_{\text{by}}$ | $Q$ | $\Delta U$ | $\Delta S_{\text{gas}}$ | Relation | $P$–$V$ cue |
|---|---|---|---|---|---|---|
| Isochoric ($V$ const.) | 0 | $n\bar{C}_v\Delta T$ | $n\bar{C}_v\Delta T$ | $n\bar{C}_v \ln \frac{T_2}{T_1}$ | $P/T$ const. | vertical line |
| Isobaric ($P$ const.) | $P\Delta V = nR\Delta T$ | $n\bar{C}_p\Delta T$ | $n\bar{C}_v\Delta T$ | $n\bar{C}_p \ln \frac{T_2}{T_1}$ | $V/T$ const. | horizontal line |
| Isothermal ($T$ const.) | $nRT \ln \frac{V_2}{V_1}$ | $= W_{\text{by}}$ | 0 | $nR \ln \frac{V_2}{V_1}$ | $PV$ const. | hyperbola, slope $-P/V$ |
| Adiabatic ($Q = 0$, reversible) | $\frac{P_1V_1 - P_2V_2}{\gamma - 1} = -n\bar{C}_v\Delta T$ | 0 | $n\bar{C}_v\Delta T$ | 0 | $PV^\gamma$, $TV^{\gamma-1}$ const. | steeper: $-\gamma P/V$ |
| Polytropic ($PV^{n_p}$) | $\frac{P_2V_2 - P_1V_1}{1 - n_p}$ | $n\bar{C}_v \frac{\gamma - n_p}{1 - n_p}\Delta T$ | $n\bar{C}_v\Delta T$ | $n\bar{C}_v \frac{\gamma - n_p}{1 - n_p} \ln \frac{T_2}{T_1}$ | $TV^{n_p - 1}$ const. | between isotherm and adiabat for $1 < n_p < \gamma$ |
| Free expansion (into vacuum; irreversible) | 0 | 0 | 0 ($T$ unchanged) | $nR \ln \frac{V_2}{V_1} > 0$ (all generated) | $P_1V_1 = P_2V_2$ | endpoints only (no quasistatic path) |
| Complete cycle | enclosed area (clockwise +) | $Q_{\text{net}} = W_{\text{net}}$ | 0 | 0 for gas; $S_{\text{gen}} \geq 0$ in reservoirs | returns to start | closed loop |

## 1.4 Units, signs, and conventions

**Table 1.3.** Convention map used in this binder.

| Item | Convention |
|---|---|
| First law (default) | $\Delta U = Q - W_{\text{by}}$; $Q > 0$ into system |
| Chemistry form | $\Delta U = q + w$, $w = W_{\text{on}} = -W_{\text{by}}$ |
| Cycles | $Q_H, Q_L, W$ positive magnitudes; arrows give direction |
| Reaction | $\Delta_r H < 0$ exothermic; per mole of reaction as written |
| Entropy balance | heats into the system positive; $S_{\text{gen}} \geq 0$ |
| Heat transfer | $\dot{Q} > 0$ from hot to cold along the path |
| Standard state | $p^\circ = 1\,\text{bar}$; tables at $298.15\,\text{K}$ |
| Pressure | absolute unless "gauge" is stated |

**KEY IDEA / DECISION Kelvin required** in: $PV = nRT$; any $T_2/T_1$ or $\ln T$; Carnot; $\sigma T^4$ and $h_r$; $\Delta G = \Delta H - T\Delta S$; $RT \ln K$; Clausius–Clapeyron; kinetic speeds. **Celsius allowed** only for differences: $mc\Delta T$, $hA\Delta T$, $kA\Delta T/L$, $T - T_\infty$ in cooling laws.

**Unit hazards that cost points**

- kJ vs J (heats of formation in kJ/mol, $R$ in $\text{J/(mol K)}$).
- g vs kg (specific heat per gram vs per kilogram; molar mass in kg/mol for speeds).
- L vs $\text{m}^3$ ($1\,\text{L} = 10^{-3}\,\text{m}^3$; $1\,\text{L}\cdot\text{kPa} = 1\,\text{J}$; $1\,\text{L}\cdot\text{atm} = 101.325\,\text{J}$).
- kPa vs Pa vs atm vs bar (Tab. 7.4).
- Flux ($\text{W/m}^2$) vs rate (W); per-area R-value vs total resistance (K/W).
- Thermochemical vs IT calorie; food Calorie = kcal.
- Specific (per kg) vs molar (per mol) vs total (per object) heat capacities.

**Dimension checks:** $[RT] = \text{J/mol}$; $[PV] = \text{J}$; $[hA] = [kA/L] = \text{W/K}$; $[\sigma T^4] = \text{W/m}^2$; $[C/K] = \text{s}$; $[\Delta H/R] = \text{K}$ (Clausius–Clapeyron, van 't Hoff slopes). Logarithms and exponentials need dimensionless arguments.

**Prefix and symbol collisions:** $\sigma$ (Stefan–Boltzmann) vs $\sigma_{\text{th}}$ (stress); $k$ (Boltzmann) vs $k$ (conductivity); $n$ (moles) vs $n_N$ (number density) vs $n_p$ (polytropic index); $\alpha$ (expansion, diffusivity, absorptivity); $\nu$ (kinematic viscosity, stoichiometric coefficient, frequency). Read the context.

**Most-used constants** (full list Tab. 7.3):

- $R = 8.314\,\text{J/(mol}\cdot\text{K)} = 0.082\,06\,\text{L}\cdot\text{atm/(mol}\cdot\text{K)}$
- $k = 1.380\,649 \times 10^{-23}\,\text{J/K}$; $N_A = 6.022\,140\,76 \times 10^{23}\,\text{mol}^{-1}$
- $\sigma = 5.670 \times 10^{-8}\,\text{W/(m}^2\cdot\text{K}^4\text{)}$; $b = 2.898 \times 10^{-3}\,\text{m}\cdot\text{K}$
- $1\,\text{atm} = 101.325\,\text{kPa}$; $c_{\text{water}} = 4.18\,\text{J/(g}\cdot\text{K)}$

# TAB 2 Foundations, Calorimetry & Phases

## 2.1 Foundations: systems, temperature, thermometry, expansion

### 2.1.1 Systems and vocabulary

**[CORE]** **Cue:** "isolated", "state function", "steady", "reversible". Name the system boundary *first*; every sign below depends on it.

**Table 2.1.** System types and paired concepts.

| Term | Meaning / discriminating test |
|---|---|
| Isolated | No mass, heat, or work crosses. $\Delta U = 0$; $\Delta S \geq 0$. |
| Closed (control mass) | Energy crosses, mass does not. Default for $\Delta U = Q - W_{\text{by}}$. |
| Open (control volume) | Mass and energy cross; use flow balance (3.29) p.17. |
| Adiabatic | $Q = 0$. Temperature *can* change (work does it). Not "isothermal". |
| Diathermal | Wall that permits heat transfer. |
| State function | Depends only on state: $T, P, V, U, H, S, A, G$. $\oint \mathrm{d}X = 0$. |
| Path function | $Q$, $W$: depend on the process (inexact $\delta Q, \delta W$). |
| Intensive | Independent of amount: $T, P, \rho$, specific $u$, molar $\bar{c}_p$. |
| Extensive | Scales with amount: $m, V, U, H, S, C$. Specific = extensive/$m$. |
| Equilibrium | No unbalanced potentials (thermal, mechanical, phase, chemical); isolated system would not change. |
| Steady state | Properties constant in time *while flows persist* (a wall conducting a constant $\dot{Q}$). Not equilibrium. |
| Quasistatic | Infinitely slow sequence of equilibrium states. |
| Reversible | Quasistatic *and* no dissipation (no friction, no finite-$\Delta T$ heat transfer, no unrestrained expansion, no mixing). $S_{\text{gen}} = 0$. |

**KEY IDEA / DECISION** Heat and work are energy *in transit* across a boundary; a body stores internal energy $U$, not "heat". Heat is transfer driven by a temperature difference; work is every other boundary energy transfer (shaft, boundary motion, electrical).

**Zeroth law.** If $A$ and $B$ are each in thermal equilibrium with $C$, then $A$ and $B$ are in thermal equilibrium with each other. This makes "temperature" a well-defined property and justifies thermometers (the thermometer is body $C$).[EK]

### 2.1.2 Temperature scales and the kelvin

**[CORE]**

**FORMULA**

$$T/\mathrm{K} = t/^\circ\mathrm{C} + 273.15 \tag{2.1}$$

$$t/^\circ\mathrm{F} = 1.8\,(t/^\circ\mathrm{C}) + 32 \tag{2.2}$$

$$T/^\circ\mathrm{R} = t/^\circ\mathrm{F} + 459.67 = 1.8\,(T/\mathrm{K}) \tag{2.3}$$

$$\Delta T\,(\mathrm{K}) = \Delta t\,(^\circ\mathrm{C}); \quad \Delta t\,(^\circ\mathrm{F}) = \Delta T\,(^\circ\mathrm{R}) = 1.8\,\Delta T\,(\mathrm{K}) \tag{2.4}$$

Anchors: $0\,\mathrm{K} = -273.15\,^\circ\mathrm{C} = -459.67\,^\circ\mathrm{F}$; $-40\,^\circ\mathrm{C} = -40\,^\circ\mathrm{F}$; water triple point $273.16\,\mathrm{K} = 0.01\,^\circ\mathrm{C}$, $611.65\,\mathrm{Pa}$.[IAPWS] Water boils at $99.97\,^\circ\mathrm{C}$ at $101.325\,\mathrm{kPa}$ on today's scale (ITS-90/IAPWS-95), not exactly $100\,^\circ\mathrm{C}$.[IAPWS]

**Absolute vs interval.** Use kelvin (or rankine) whenever $T$ appears *multiplied, divided, in a ratio, a power, or a logarithm*: $PV = nRT$, $T_2/T_1$, Carnot, $\sigma T^4$, $\ln(T_2/T_1)$, $e^{-E/RT}$. Celsius is acceptable only for differences: $Q = mc\Delta T$, $\dot{Q} = hA\Delta T$, $kA\Delta T/L$.

**Kelvin since 20 May 2019.** The kelvin is defined by fixing the Boltzmann constant at exactly $k = 1.380\,649 \times 10^{-23}\,\mathrm{J/K}$.[CODATA] The water triple point is therefore now a *measured* value (still $273.16\,\mathrm{K}$ to within about $0.1\,\mathrm{mK}$).[EK] Because $N_A$ is also exact, $R = N_A k$ is exact.

**Rankine** is the absolute scale with Fahrenheit-sized degrees. *Historical one-liner:* Réaumur scale has 0°Ré at the ice point and 80°Ré at the boiling point (1°Ré = $1.25\,^\circ\mathrm{C}$ interval).[EK]

### 2.1.3 Thermometers

**[CORE]** Every thermometer reports *its own sensing element's* temperature (or, for IR, the radiation leaving a surface). Good practice: good thermal contact, immersion, shielding from radiation, waiting $\gtrsim$ 3–5 sensor time constants (p. 27), and calibration against fixed points.

**Table 2.2.** Thermometer comparison (qualitative; ranges and accuracies depend on the specific instrument, so none are quoted).[EK]

| Type | Physical quantity sensed | Strengths | Limitations, corrections, calibration |
|---|---|---|---|
| Liquid-in-glass | Expansion of liquid relative to glass bulb/stem | No power; simple; direct reading | Slow; immersion-depth (emergent stem) correction; parallax; fragile; Hg toxicity; reads its own bulb temperature |
| Bimetallic | Bending of two bonded strips with different $\alpha$ | Rugged, cheap; drives switches (thermostats) | Coarse accuracy; hysteresis; mechanical drift |
| Constant-volume gas | $P \propto T$ for dilute gas at fixed $V$ | Primary (absolute) thermometry; extrapolate to $P \to 0$ | Bulky, slow; corrections for dead volume and bulb expansion |
| Thermocouple | Seebeck emf set by temperature *difference* between measuring and reference junctions | Wide range; tiny; fast; cheap | Needs cold-junction compensation; µV/K-scale signal; nonlinear; type-specific tables; inhomogeneity errors |
| RTD (Pt) | Electrical resistance of platinum (rises nearly linearly with $T$) | Accurate, stable, repeatable | Self-heating from current; lead resistance (use 3- or 4-wire); slower, costlier |
| Thermistor | Resistance of semiconductor (NTC: falls steeply with $T$) | High sensitivity; small | Strongly nonlinear (fit, e.g. Steinhart–Hart $1/T = A + B\ln R + C(\ln R)^3$); limited range; self-heating |
| Infrared (radiation) | Thermal radiation from a surface spot | Non-contact; fast; moving or hot targets | Needs correct emissivity; shiny surfaces read low; sees through nothing opaque to IR (e.g. reads glass surface, not water inside); spot-size averaging; reflections |

**Two-point linear calibration** (reading $r$, reference temperatures $T_a, T_b$ with readings $r_a, r_b$):

$$T = T_a + (T_b - T_a)\,\frac{r - r_a}{r_b - r_a}. \tag{2.5}$$

Ice point: stirred slurry of crushed ice and pure water, 0.00 °C at 1 atm. Boiling point: depends on pressure—use Tab. 2.8 (p. 13), not 100 °C. Calibrate over the range you will use; linear interpolation between two points does not fix nonlinearity (thermistors).

**TRAP / LIMIT** A thermometer in air near a hot lamp or a cold window reads a mix of air convection and *radiation* exchange; shield it. An IR gun aimed at a shiny can or a glass beaker reports a meaningless number unless emissivity is set for that surface.

### 2.1.4 Thermal expansion

**[CORE] Cue:** rails, rings over shafts, overflowing liquid, bimetal, stress in a clamped bar.

**FORMULA**

$$\Delta L = \alpha L_0\,\Delta T, \qquad \Delta A \approx 2\alpha A_0 \Delta T, \tag{2.6}$$

$$\Delta V = \beta V_0\,\Delta T, \qquad \beta \approx 3\alpha \text{ (isotropic solid)}. \tag{2.7}$$

Assumptions: small strain ($\alpha\Delta T \ll 1$), $\alpha$ constant over the range (otherwise $\ln(L/L_0) = \int \alpha\,\mathrm{d}T$), isotropic material. A **hole** in a plate grows exactly as if it were filled with the same material.

**Apparent expansion** of a liquid in a container that also expands:

$$\Delta V_{\text{overflow}} = V_0\,(\beta_{\text{liq}} - 3\alpha_{\text{cont}})\,\Delta T. \tag{2.8}$$

**Thermal stress** in a bar fully restrained at both ends (linear elastic, uniaxial, no buckling or yielding):

$$\sigma_{\text{th}} = -E\,\alpha\,\Delta T \quad \text{(negative = compressive on heating)}. \tag{2.9}$$

Notation: $\sigma_{\text{th}}$ is mechanical stress (Pa); do not confuse with the Stefan–Boltzmann constant $\sigma$ in (5.17) p.26.

**SOLVED DEMONSTRATION Restrained rail.** Steel, $E = 200\,\text{GPa}$, $\alpha = 12 \times 10^{-6}\,\text{K}^{-1}$ (given values), heated 40 K with ends fixed: $\sigma_{\text{th}} = -(200\times10^9)(12\times10^{-6})(40) = -96\,\text{MPa}$ (compressive). The length $L$ cancels: stress does not depend on bar length; the *free* elongation would.

**Bimetal strip** bends with the higher-$\alpha$ metal on the *outside* (convex side) when heated. **Ideal gas** at constant $P$: $\beta = 1/T$ (absolute $T$).

**Water's density anomaly.** At 101.325 kPa, liquid water is densest at 3.98 °C (999.97 kg/m$^3$); $\beta < 0$ from 0 to 3.98 °C.[IAPWS] Ice Ih at 0 °C has $\rho = 916.7\,\text{kg/m}^3$ versus $999.84\,\text{kg/m}^3$ for the liquid, so freezing expands volume by about 9.1 %.[ICE] Consequences: ice floats; lakes stratify with 4 °C water at depth in winter and freeze from the top; freezing water bursts pipes.

**Table 2.3.** Representative coefficients near room temperature. Composition- and grade-dependent: **use the value a question supplies**.

| Material | $\alpha$ or $\beta$ ($\mathrm{K}^{-1}$) |
|---|---|
| Aluminum ($\alpha$) | $\approx 23 \times 10^{-6}$[EK] |
| Copper / brass ($\alpha$) | $\approx 17 \times 10^{-6}$ / $\approx 19 \times 10^{-6}$[EK] |
| Carbon steel ($\alpha$) | $\approx 12 \times 10^{-6}$[EK] |
| Invar ($\alpha$) | $\approx 1.2 \times 10^{-6}$[EK] |
| Soda-lime / borosilicate glass ($\alpha$) | $\approx 9 \times 10^{-6}$ / $\approx 3.3 \times 10^{-6}$[EK] |
| Water at 20 °C ($\beta$) | $2.07 \times 10^{-4}$[IAPWS] |
| Ideal gas at 300 K ($\beta = 1/T$) | $3.33 \times 10^{-3}$ |

## 2.2 Calorimetry and the phase-change ledger

### 2.2.1 Heat capacities

**[CORE] Cue:** "specific heat", "heat capacity of the cup", "water equivalent", "J/(mol K)".

**Table 2.4.** Three bases—keep the basis visible in the units.

| Symbol | Name | Units |
|---|---|---|
| $C$ | total heat capacity of an object | J/K |
| $c = C/m$ | specific (mass) heat capacity | J/(g · K), J/(kg · K) |
| $\bar{c} = C/n$ | molar heat capacity | J/(mol · K) |

**FORMULA**

$$Q = \int_{T_1}^{T_2} C(T)\,\mathrm{d}T \xrightarrow{C \text{ const}} Q = mc\,\Delta T \tag{2.10}$$

$$c_v = \left(\frac{\partial u}{\partial T}\right)_v, \qquad c_p = \left(\frac{\partial h}{\partial T}\right)_p \tag{2.11}$$

**Which one?** Closed system, rigid, no work: $Q_V = \Delta U$. Closed system at constant pressure with only boundary work: $Q_P = \Delta H$. Solids and liquids: $c_p \approx c_v$ (tables list $c_p$); gases: $c_p - c_v = R$ per mole ((3.12) p.15).

**Constant-$c$ shortcut is conditional.** Water's $c_p$ varies only ±0.6 % between 15 and 90 °C (4.188 to 4.181 to 4.205 J/(g·K)),[IAPWS] so $c = 4.18\,\mathrm{J/(g\cdot K)}$ is fine; but use the integral (or tabulated $\Delta h$) for wide ranges, for gases over hundreds of kelvin, and near phase changes.

**Water equivalent** of a container: $m_{\mathrm{eq}} = C_{\mathrm{cal}}/c_{\mathrm{water}}$ (grams of water that would absorb the same energy per kelvin).

**Dulong–Petit check** (many solid elements near room temperature): $\bar{c}_p \approx 3R = 24.9\,\mathrm{J/(mol\cdot K)}$, so an unknown metal's molar mass $M \approx 3R/c$. Tab. 7.2 shows 24.2–26.4. Fails for light, stiff solids (diamond, Be, B) and at low $T$.[CRC][EK]

### 2.2.2 Water ledger constants

**Table 2.5.** Water, ice, steam at 101.325 kPa unless stated.[IAPWS][ICE]

| Quantity | Value |
|---|---|
| $c_p$ liquid, 25 °C (15 / 90 °C) | 4.181 J/(g · K) (4.188 / 4.205) |
| $c_p$ ice Ih at 0 / −10 / −20 °C | 2.097 / 2.023 / 1.950 J/(g · K) |
| $c_p$ steam: sat. vapor / 120 °C / 150 °C | 2.080 / 2.021 / 1.986 J/(g · K) |
| $L_f$ (fusion) at 0 °C | 333.4 J/g (6.007 kJ/mol) |
| $L_v$ at normal boiling point 99.97 °C | 2256.5 J/g (40.65 kJ/mol) |
| $L_v$ at 25 °C (evaporation, at $p_{\mathrm{sat}}$) | 2441.7 J/g (43.99 kJ/mol) |
| $L_{\mathrm{sub}}$ near triple point $\approx L_f + L_v$(0.01 °C) | ≈ 2834 J/g |
| $\rho$ liquid 25 °C / ice 0 °C | 997.0 / 916.7 kg/m$^3$ |

Textbook roundings $c = 4.18$ or 4.184, $L_f = 334$, $L_v = 2260$ are acceptable *if a question states them*; use the given values when supplied.

### 2.2.3 The energy ledger

**[CORE] Cue:** "mixed", "dropped into", "final temperature", "unknown metal", "heater".

**FORMULA** For bodies $i$ inside an insulated, constant-pressure enclosure, with net energy input $E_{\mathrm{in}}$ (electrical, stirring) and leak $Q_{\mathrm{loss}}$:

$$\sum_i \Delta H_i \;=\; E_{\mathrm{in}} - Q_{\mathrm{loss}}, \tag{2.12}$$

each $\Delta H_i$ a sum of sensible pieces $m_i c_i (T_f - T_i)$ and latent pieces $\pm mL$ (+ for melting/boiling, − for freezing/condensing).

**No phase change, no input, no leak, constant $C$:**

$$T_f = \frac{\sum_i C_i T_i}{\sum_i C_i}, \qquad C_i = m_i c_i. \tag{2.13}$$

**KEY IDEA / DECISION Ledger algorithm.** (1) List *every* body: samples, water, cup, stirrer, probe. (2) Hypothesize the final state (temperature region and phases). (3) Write each $\Delta H_i$ with signed $(T_f - T_i)$—no "lost/gained" bookkeeping. (4) Sum to $E_{\text{in}} - Q_{\text{loss}}$ and solve. (5) Reject and redo if the result contradicts the hypothesis (e.g. $T_f > 0\,^\circ\text{C}$ while ice remains, or melted mass $< 0$ or $>$ available).

**TRAP / LIMIT** With no external input, $T_f$ must lie between the lowest and highest initial temperatures. A result outside that range signals a sign error, a missed phase change, an unaccounted reaction or electrical input, or a wrong capacity. "Heat lost = heat gained" with magnitudes invites sign errors; the signed sum cannot.

### 2.2.4 Phase-change decision procedure

**[CORE]** Ice (at $T_{\text{ice}} \le 0$) added to warm water (plus cup) at $T_w$; at 1 atm use $0\,^\circ\text{C}$ as the melting point. Define the energy terms with all capacities $> 0$:

$$\begin{aligned} Q_{\text{av}} &= (m_w c_w + C_{\text{cup}})\,(T_w - 0) && \text{warm side} \to 0\,^\circ\text{C} \\ Q_1 &= m_i c_{\text{ice}}\,(0 - T_{\text{ice}}) && \text{ice} \to 0\,^\circ\text{C} \\ Q_2 &= m_i L_f && \text{melt all ice} \end{aligned}$$

**KEY IDEA / DECISION Decide by comparing energies, never by guessing $T_f$:**

1. $Q_{\text{av}} < Q_1$: some *water freezes*; frozen mass $= (Q_1 - Q_{\text{av}})/L_f$. If that exceeds $m_w$, everything freezes and $T_f < 0\,^\circ\text{C}$ (solve sensible ledger).
2. $Q_1 \le Q_{\text{av}} < Q_1 + Q_2$: $T_f = 0\,^\circ\text{C}$; melted mass $= (Q_{\text{av}} - Q_1)/L_f$.
3. $Q_{\text{av}} \ge Q_1 + Q_2$: all melts; $T_f = \dfrac{Q_{\text{av}} - Q_1 - Q_2}{(m_w + m_i)c_w + C_{\text{cup}}}$.

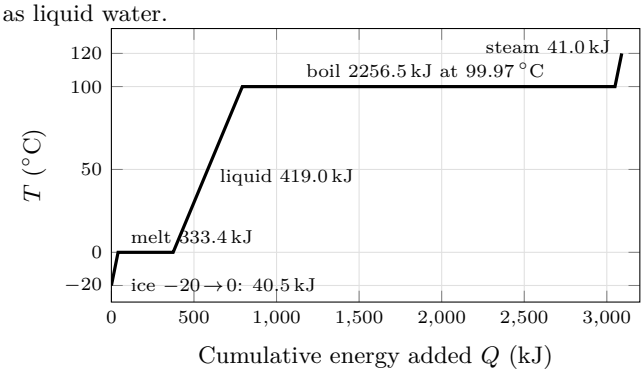
Steam (at $T_{\text{sat}}$) into cool water: same logic at $T_{\text{sat}}(P)$ (Tab. 2.8), with condensation energy $m_s L_v$ released first.

**SOLVED DEMONSTRATION Partial melt.** 100.0 g ice at $-10.0\,^\circ\text{C}$ into 200.0 g water at $25.0\,^\circ\text{C}$ in a cup with $C_{\text{cup}} = 10\,\text{J/K}$; $c_w = 4.18$, $c_{\text{ice}} = 2.03\,\text{J/(g}\cdot\text{K)}$, $L_f = 333.4\,\text{J/g}$. $Q_{\text{av}} = (836 + 10)(25) = 21\,150\,\text{J}$; $Q_1 = 100(2.03)(10) = 2030\,\text{J}$; $Q_2 = 33\,340\,\text{J}$. Case 2 applies $(2030 \le 21150 < 35370)$: $T_f = 0\,^\circ\text{C}$, melted $= (21150 - 2030)/333.4 = 57.35\,\text{g}$, so 42.65 g of ice remains. *Check:* signed ledger $-21150 + 2030 + 57.35(333.4) = 0$ ✓; $0 < 57.35 < 100$ ✓.

**SOLVED DEMONSTRATION Steam condenses.** 10.0 g steam at $100\,^\circ\text{C}$ into 200.0 g water at $20.0\,^\circ\text{C}$ (ignore cup; $L_v = 2256.5\,\text{J/g}$, $c_w = 4.18$): $10(-2256.5) + 10(4.18)(T_f - 100) + 200(4.18)(T_f - 20) = 0 \Rightarrow T_f = 49.5\,^\circ\text{C}$. Only 10 g of steam raises 200 g of water by 29.5 K: latent energy dominates.

### 2.2.5 Heating curve

**Figure 2.1.** Heating curve of 1.00 kg $H_2O$ at 101.325 kPa, drawn to scale from IAPWS enthalpy differences.[IAPWS][ICE] Plateaus: constant $T$ while phase changes. Sloped segments: slope $= 1/(mc_p)$, so ice and steam rise about twice as steeply as liquid water.




### 2.2.6 Calorimeters and corrections

**[CORE]**

**FORMULA**

Coffee cup (constant $P$), $\Delta H \approx q_{\text{rxn}}$ :

$$q_{\text{rxn}} = -(m_{\text{soln}} c_{\text{soln}} + C_{\text{cal}})\,\Delta T \tag{2.14}$$

Bomb (constant $V$), $\Delta U \approx q_{\text{rxn}}$ :

$$q_{\text{rxn}} = -C_{\text{bomb}}\,\Delta T \tag{2.15}$$

Electrical: $E_{\text{in}} = VIt = I^2Rt = \frac{V^2}{R}\,t$ (2.16)

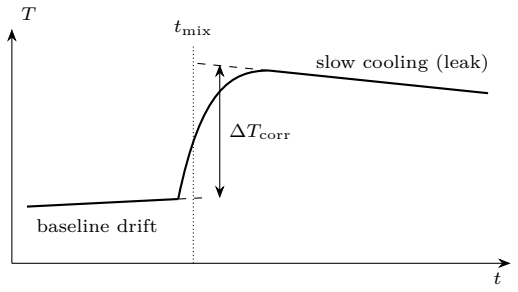
Divide $q_{\text{rxn}}$ by moles of the *limiting* reactant for a molar $\Delta H$; then convert bomb $\Delta U$ to $\Delta H$ with (6.4) p.29.

**Coffee-cup assumptions:** negligible exchange with the room during the run (foam cup, lid), solution $c$ and density ≈ water's (often $4.18\,\text{J/(g}\cdot\text{K)}$, 1.00 g/mL), complete reaction, no gas or vapor loss. **Bomb:** calibrated $C_{\text{bomb}}$ from a standard (benzoic acid); corrections for ignition-wire energy, acid formation, and jacket exchange.[EK] Neither is lossless.

**Heater energy ≠ sample energy.** $E_{\text{in}}$ is split among sample, container, heater, probe, and losses. Electrical calibration: $C_{\text{total}} = E_{\text{in}}/\Delta T$ for the whole assembly.

**Heat-leak (drift) correction.** Record $T$ before and after mixing; extrapolate both linear (or exponential) drift lines to the moment of mixing and take $\Delta T$ between the extrapolations. This approximately removes exchange with the room during slow mixing.[EK]

**Figure 2.2.** Heat-leak (drift) correction, schematic (axes not to scale). Extrapolate the pre- and post-event drift lines to the mixing time $t_{\text{mix}}$; use $\Delta T$ between them, not the raw peak.[EK]




**SOLVED DEMONSTRATION Electrical calibration.** A heater at 12.0 V, 1.50 A for 300 s delivers $E = 5400$ J to a cup holding 250 g water; drift-corrected $\Delta T = 4.95$ K. $C_{\text{total}} = 5400/4.95 = 1091$ J/K; minus water ($250 \times 4.18 = 1045$ J/K) gives $C_{\text{cal}} \approx 46$ J/K. Small differences of large numbers: a 0.05 K error in $\Delta T$ shifts $C_{\text{cal}}$ by about 11 J/K.[DER]

**SOLVED DEMONSTRATION Calorimeter constant.** 50.0 g water at 60.0 °C poured into a cup holding 50.0 g water at 20.0 °C; $T_f = 38.6$ °C; $c = 4.184$ J/(g · K). Hot water: $50(4.184)(38.6 - 60.0) = -4477$ J. Cold water: $50(4.184)(18.6) = 3891$ J. Cup: +586 J over 18.6 K $\Rightarrow C_{\text{cal}} = 31.5$ J/K (water equivalent 7.5 g). *Check:* resolving (2.12) with $C_{\text{cal}} = 31.49$ returns $T_f = 38.60$ °C ✓.

**SOLVED DEMONSTRATION Unknown metal, and why the cup matters.** 100.0 g metal at 98.0 °C into 150.0 g water at 22.0 °C in that cup ($C_{\text{cal}} = 31.5$ J/K); $T_f = 26.2$ °C. $c_m = \dfrac{(150 \cdot 4.184 + 31.5)(4.2)}{100\,(71.8)} = 0.386$ J/(g · K). Ignoring the cup gives 0.367 (5 % low). *Resolution:* ±0.1 K on $T_f$ gives 0.376–0.395, which contains Cu (0.384) *and* Zn (0.388) but excludes Fe, Al, Ni (Tab. 7.2). Dulong–Petit: $M \approx 24.9/0.386 = 65$ g/mol—still Cu or Zn. A second property (density) is needed. Report the range, not a false identification.

**TRAP / LIMIT** Common ledger errors: forgetting the cup/probe; using 100 °C for boiling at altitude; using $L_v$ at 100 °C for room-temperature evaporation (2441.7 J/g at 25 °C); using $c_{\text{water}}$ for ice or steam; mixing grams with kilograms or J with cal ($1\,\text{cal}_{\text{th}} = 4.184$ J; food Calorie = 1 kcal); using a molar $\bar{c}$ with a mass.

## 2.3 Phases, vapor pressure, and humidity

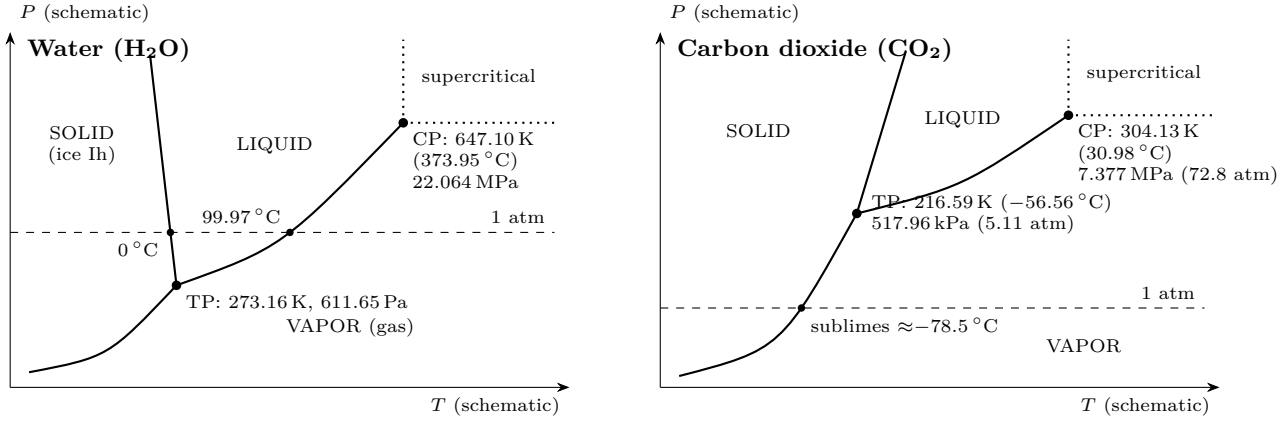
### 2.3.1 Transitions: latent vs sensible

**[CORE] Sensible** energy changes temperature within one phase; **latent** energy changes phase at constant $T$ and $P$. Toward the higher-entropy phase, $\Delta H > 0$ (absorbed).

**Table 2.6.** Transition names and signs.

| From → to | Name | $\Delta H$ |
|---|---|---|
| solid → liquid | melting (fusion) | + |
| liquid → solid | freezing (solidification) | − |
| liquid → gas | vaporization: *boiling* (bubbles form in the bulk when $p_{\text{sat}}(T) = P_{\text{ext}}$) or *evaporation* (surface, any $T$) | + |
| gas → liquid | condensation | − |
| solid → gas | sublimation | + |
| gas → solid | deposition (desublimation) | − |

**Figure 2.3.** Schematic $P$–$T$ phase diagrams (**not to scale**; pressure axis compressed, log-like). Key coordinates are quantitative.[IAPWS][CP] Water's fusion line slopes *left* (ice is less dense than liquid); $CO_2$'s slopes right, and its triple point lies above 1 atm, so solid $CO_2$ sublimes at 1 atm (near −78.5 °C).[EK]



**Reading the diagram.** Horizontal line = isobaric heating; vertical line = isothermal compression. Each curve is two-phase coexistence ($F = 1$); TP: three phases ($F = 0$). Above the critical point there is no liquid–vapor boundary: a path around CP converts liquid to gas continuously, and $h_{fg} \to 0$ at CP. Compressing ice just below 0 °C can melt it because water's fusion line slopes left (but only by 7.5 mK per atm, below).

**TRAP / LIMIT** Pressure melting alone does not explain skating: at typical rink temperatures the needed pressures are far larger than 13.5 MPa/K allows for, and surface premelting and frictional heating are invoked; the mechanism is still discussed. Likewise the Mpemba effect (hot water "freezing faster") is condition-dependent and not a law, and the Leidenfrost temperature depends on the liquid, surface, and roughness.[EK]

### 2.3.2 Phase rule and quality

**[CORE]**

**FORMULA**

$$F = C - P_{\mathrm{h}} + 2 \tag{2.17}$$

$$x = \frac{m_{\mathrm{vap}}}{m_{\mathrm{liq}} + m_{\mathrm{vap}}}, \quad y = y_f + x\,(y_g - y_f) \tag{2.18}$$

$F$ = independent intensive variables, $C$ = independent components, $P_{\mathrm{h}}$ = number of phases. Limits: assumes only $T$ and $P$ as field variables, no reactions (each independent reaction lowers $C$ by one), no surface/electric/magnetic work. It gives *how many* variables are free, not the amounts present.

Quality $x$ is a **mass** fraction, defined only in the two-phase dome ($0 \le x \le 1$); $y$ is any specific property ($v, u, h, s$). Lever rule: $x = (v - v_f)/(v_g - v_f)$. Compressed liquid and superheated vapor have no quality. Even $x = 0.01$ is mostly vapor *by volume* at low pressure ($v_g/v_f \approx 1600$ at 1 atm).

### 2.3.3 Clapeyron and Clausius–Clapeyron

**[CORE]**

**FORMULA** Exact along any two-phase coexistence curve:

$$\frac{\mathrm{d}P}{\mathrm{d}T} = \frac{\Delta h}{T\,\Delta v} = \frac{\Delta s}{\Delta v}. \tag{2.19}$$

Approximate (Clausius–Clapeyron) for vaporization or sublimation:

$$\ln\frac{P_2}{P_1} = -\frac{\Delta \bar{h}_{\mathrm{vap}}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right). \tag{2.20}$$

Assumptions of (2.20): $v_g \gg v_f$; vapor ideal; $\Delta\bar{h}$ constant (molar, J/mol, with $R = 8.314\,\mathrm{J/(mol\cdot K)}$; or specific with $R_s = R/M$). Use kelvin. A plot of $\ln P$ vs $1/T$ has slope $-\Delta\bar{h}/R$. Valid over modest ranges well below the critical point. Sign: vaporization and sublimation always have $\mathrm{d}P/\mathrm{d}T > 0$; melting has the sign of $\Delta v$ (negative for water).

**SOLVED DEMONSTRATION Boiling at reduced pressure; validity limit.** Water at 84.0 kPa: with $\Delta\bar{h}_{\mathrm{vap}} = 40.65$ kJ/mol at 373.12 K, (2.20) gives $T_2 = 367.86\,\mathrm{K} = 94.71\,^\circ\mathrm{C}$; IAPWS-95 gives 94.80 °C (error 0.09 K). *Extrapolating far* (from 100 °C down to 20 °C) gives 2.84 kPa versus 2.34 kPa (21 % high), because $\Delta\bar{h}_{\mathrm{vap}}$ rises to 44.2 kJ/mol at 20 °C.[IAPWS]

**SOLVED DEMONSTRATION How much does pressure lower ice's melting point?** $\Delta h = 333.4$ kJ/kg, $\Delta v = v_{\mathrm{liq}} - v_{\mathrm{ice}} = -9.07 \times 10^{-5}\,\mathrm{m^3/kg}$ at 273.15 K: $\mathrm{d}P/\mathrm{d}T = 333400/(273.15 \times -9.07 \times 10^{-5}) = -13.5$ MPa/K, i.e. −7.5 mK per atm. *Check:* the IAPWS ice-Ih melting curve gives −13.45 MPa/K.[ICE]

**Antoine equation** $\log_{10} P = A - B/(C + T)$: every coefficient set is tied to specific pressure units, temperature units (°C or K), logarithm base, and a validity window. Never mix sets or extrapolate outside the window. (No Antoine table is included: no coefficient set could be source-verified for this build; use Tab. 2.7 for water.)

### 2.3.4 Boiling, evaporation, and metastability

**Boiling** begins when $p_{\mathrm{sat}}(T)$ reaches the local pressure: lower at altitude, higher in a pressure cooker (200 kPa absolute → 120.2 °C).[IAPWS] **Evaporation** occurs at any temperature when the surrounding vapor pressure is below $p_{\mathrm{sat}}(T_{\mathrm{surface}})$; it removes $h_{fg}$ per unit mass from the liquid (*evaporative cooling*).

**SOLVED DEMONSTRATION Evaporation from an open beaker.** At 60 °C, $h_{fg} = 2357.7$ J/g. Evaporating just 1.0 g removes enough energy to cool 100 g of water by $2357.7/(100 \times 4.18) = 5.6$ K. A lid often matters more than extra wall insulation.[IAPWS]

**Metastable states:** supercooled liquid (below freezing point), superheated liquid (above boiling point), supersaturated vapor. They persist without nucleation sites; when nucleation occurs, the latent energy released drives the mixture back toward the coexistence temperature (apply the ledger). **Supercritical fluid** ($T > T_c$, $P > P_c$): single phase with gas-like and liquid-like properties; no surface tension or $h_{fg}$.

### 2.3.5 Humidity, dew point, gas over water

**[CORE]**

**FORMULA**

$$\phi = \frac{p_v}{p_{\mathrm{sat}}(T)}, \qquad p_{\mathrm{sat}}(T_{\mathrm{dp}}) = p_v \tag{2.21}$$

$$\omega = \frac{M_w}{M_a}\,\frac{p_v}{P - p_v} = 0.622\,\frac{p_v}{P - p_v} \tag{2.22}$$

$$P_{\mathrm{gas}} = P_{\mathrm{atm}} - p_{\mathrm{sat,w}}(T) \quad \text{(levels equalized)} \tag{2.23}$$

$\phi$ = relative humidity; $T_{\mathrm{dp}}$ = dew point: temperature at which air cooled at constant pressure and constant water content becomes saturated; $\omega$ = humidity ratio (kg water per kg dry air; $0.622 = 18.015/28.965$). Below 0 °C use the ice curve (*frost point*, Tab. 2.9). In (2.23), if the water level inside the collection tube is higher than outside by $h$, subtract $\rho g h$ as well.

**Dew point ≠ wet bulb.** The wet-bulb temperature is

set by evaporative cooling of a wetted sensor in moving air (a heat–mass balance). Always $T_{\mathrm{dp}} \le T_{\mathrm{wb}} \le T_{\mathrm{db}}$, with equality only at 100 % RH.[EK]

**SOLVED DEMONSTRATION Dew point.** Air at 25 °C, $\phi = 60\,\%$, 1 atm. $p_{\mathrm{sat}}(25\,^\circ\mathrm{C}) = 3.170\,\mathrm{kPa}$, so $p_v = 1.902\,\mathrm{kPa}$. Solve $p_{\mathrm{sat}}(T_{\mathrm{dp}}) = 1.902$: $T_{\mathrm{dp}} = 16.7\,^\circ\mathrm{C}$ (linear interpolation between 15 °C and 20 °C rows gives 16.55 °C, slightly low because $p_{\mathrm{sat}}(T)$ is convex). $\omega = 0.622(1.902)/(101.325 - 1.902) = 0.0119\,\mathrm{kg/kg}$.

**Table 2.7.** Saturated water by temperature (IAPWS-95).[IAPWS] Reference state: $u = s = 0$ for saturated liquid at the triple point; only *differences* of $h$ and $s$ are physical.

| $T$ (°C) | $p_{\mathrm{sat}}$ (kPa) | $\rho_f$ (kg/m$^3$) | $v_g$ (m$^3$/kg) | $h_f$ (kJ/kg) | $h_{fg}$ (kJ/kg) | $h_g$ (kJ/kg) | $s_f$ (kJ/(kg · K)) | $s_g$ (kJ/(kg · K)) |
|---|---|---|---|---|---|---|---|---|
| 0.01 | 0.6117 | 999.8 | 206.0 | 0.0 | 2500.9 | 2500.9 | 0.0000 | 9.1555 |
| 5 | 0.8726 | 999.9 | 147.0 | 21.0 | 2489.0 | 2510.1 | 0.0763 | 9.0248 |
| 10 | 1.228 | 999.7 | 106.3 | 42.0 | 2477.2 | 2519.2 | 0.1511 | 8.8998 |
| 15 | 1.706 | 999.1 | 77.88 | 63.0 | 2465.4 | 2528.3 | 0.2245 | 8.7803 |
| 20 | 2.339 | 998.2 | 57.76 | 83.9 | 2453.5 | 2537.4 | 0.2965 | 8.6660 |
| 25 | 3.170 | 997.0 | 43.34 | 104.8 | 2441.7 | 2546.5 | 0.3672 | 8.5566 |
| 30 | 4.247 | 995.6 | 32.88 | 125.7 | 2429.8 | 2555.5 | 0.4368 | 8.4520 |
| 35 | 5.629 | 994.0 | 25.21 | 146.6 | 2417.9 | 2564.5 | 0.5051 | 8.3517 |
| 40 | 7.385 | 992.2 | 19.52 | 167.5 | 2406.0 | 2573.5 | 0.5724 | 8.2555 |
| 50 | 12.35 | 988.0 | 12.03 | 209.3 | 2381.9 | 2591.3 | 0.7038 | 8.0748 |
| 60 | 19.95 | 983.2 | 7.667 | 251.2 | 2357.7 | 2608.8 | 0.8313 | 7.9081 |
| 70 | 31.20 | 977.7 | 5.040 | 293.1 | 2333.0 | 2626.1 | 0.9551 | 7.7540 |
| 80 | 47.41 | 971.8 | 3.405 | 335.0 | 2308.0 | 2643.0 | 1.0756 | 7.6111 |
| 90 | 70.18 | 965.3 | 2.359 | 377.0 | 2282.5 | 2659.5 | 1.1929 | 7.4781 |
| 100 | 101.4 | 958.3 | 1.672 | 419.2 | 2256.4 | 2675.6 | 1.3072 | 7.3541 |
| 120 | 198.7 | 943.1 | 0.8912 | 503.8 | 2202.1 | 2705.9 | 1.5279 | 7.1291 |
| 150 | 476.2 | 917.0 | 0.3925 | 632.2 | 2113.7 | 2745.9 | 1.8418 | 6.8371 |

**Table 2.8.** Boiling point of water vs absolute pressure.[IAPWS]

| $P$ (kPa) | $T_{\mathrm{sat}}$ (°C) | $h_{fg}$ (kJ/kg) |
|---|---|---|
| 10 | 45.81 | 2392.1 |
| 20 | 60.06 | 2357.5 |
| 50 | 81.32 | 2304.7 |
| 70 | 89.93 | 2282.7 |
| 80 | 93.49 | 2273.5 |
| 84 | 94.80 | 2270.0 |
| 90 | 96.69 | 2265.1 |
| 101.325 | 99.97 | 2256.5 |
| 150 | 111.35 | 2226.0 |
| 200 | 120.21 | 2201.5 |
| 300 | 133.52 | 2163.5 |
| 500 | 151.83 | 2108.0 |
| 1000 | 179.88 | 2014.6 |
| 2000 | 212.38 | 1889.8 |

**Table 2.9.** Sublimation (frost-point) pressure over ice Ih.[ICE]

| $T$ (°C) | $p_{\mathrm{sub}}$ (Pa) |
|---|---|
| −30 | 38.0 |
| −20 | 103.2 |
| −10 | 259.9 |
| −5 | 401.7 |
| 0 | 611.2 |

**Use of these tables.** Saturated mixtures: $h = h_f + x h_{fg}$, $s = s_f + x(s_g - s_f)$. Interpolate linearly only between close rows; $p_{\mathrm{sat}}$ grows roughly exponentially, so prefer (2.20) between two nearby rows for an in-between pressure. Note the triple-point pressure 611.65 Pa at 0.01 °C vs 611.2 Pa over ice at 0 °C.

# TAB 3 Gases, First Law, Second & Third Laws

## 3.1 Gases: ideal, kinetic, equipartition, real

### 3.1.1 Ideal-gas law in every basis

**[CORE]**

**FORMULA**

$$PV = nRT = NkT = mR_sT, \qquad R_s = R/M \tag{3.1}$$

$$\rho = \frac{PM}{RT}, \qquad P = \rho R_s T \tag{3.2}$$

$$\frac{P_1V_1}{n_1T_1} = \frac{P_2V_2}{n_2T_2} \quad \text{(combined gas law)} \tag{3.3}$$

$P$ **absolute** (Pa), $T$ in K, $n$ in mol, $N$ molecules, $M$ in kg/mol when $R = 8.314\,\mathrm{J/(mol\cdot K)}$. $P_{\mathrm{abs}} = P_{\mathrm{gauge}} + P_{\mathrm{atm}}$. For air $R_s = 287.05\,\mathrm{J/(kg\cdot K)}$ ($M = 28.965\,\mathrm{g/mol}$).[CP] $R$ **in other units:** $8.314\,462\,618\,\mathrm{J/(mol\cdot K)} = 0.083\,144\,626\,\mathrm{L\cdot bar/(mol\cdot K)} = 0.082\,057\,37\,\mathrm{L\cdot atm/(mol\cdot K)} = 62.363\,60\,\mathrm{L\cdot Torr/(mol\cdot K)} = 1.987\,20\,\mathrm{cal/(mol\cdot K)}$ (thermochemical cal) [CODATA]

Special cases (hold the others fixed): Boyle $PV$ const $(n,T)$; Charles $V/T$ const $(n,P)$; Gay-Lussac $P/T$ const $(n,V)$; Avogadro $V/n$ const $(T,P)$.

**Table 3.1.** "Standard" conditions are conventions—state which one.[DER]

| Convention | $T$ | $P$ | $\bar{V}$ (L/mol) |
|---|---|---|---|
| IUPAC STP (since 1982) | 273.15 K | 100 kPa | 22.711 |
| Older STP | 273.15 K | 101.325 kPa | 22.414 |
| SATP | 298.15 K | 100 kPa | 24.790 |
| 25 °C, 1 atm | 298.15 K | 101.325 kPa | 24.465 |
| "NTP" (one usage) | 293.15 K | 101.325 kPa | 24.055 |

Thermochemical standard state is $p^\circ = 1\,\mathrm{bar}$ with *no* fixed temperature (data usually at 298.15 K).

### 3.1.2 Mixtures: Dalton and mole fractions

**FORMULA**

$$P = \textstyle\sum_i p_i, \quad p_i = y_iP, \quad y_i = \frac{n_i}{n} = \frac{p_i}{P} = \frac{V_i}{V} \tag{3.4}$$

$$M_{\mathrm{mix}} = \textstyle\sum_i y_iM_i \tag{3.5}$$

$V_i$ = Amagat partial volume (component alone at mixture $T,P$). Mass fraction $w_i = y_iM_i/M_{\mathrm{mix}} \neq y_i$. Dry air: $y \approx 0.781$ $N_2$, 0.209 $O_2$, 0.0093 Ar, $\approx 0.0004$ $CO_2$ (rising).[EK] Gas over water: (2.23) p.12.

**SOLVED DEMONSTRATION Gas collected over water → molar mass.** 250.0 mL of gas over water at 22.0 °C, levels equalized, $P_{\mathrm{atm}} = 99.5\,\mathrm{kPa}$; dry-gas mass 0.316 g. $p_{\mathrm{sat}}(22\,^\circ\mathrm{C}) = 2.645\,\mathrm{kPa}$,[IAPWS] so $p_{\mathrm{gas}} = 96.85\,\mathrm{kPa}$; $n = pV/RT = 9.867\,\mathrm{mmol}$; $M = 32.0\,\mathrm{g/mol}$ ($O_2$). Ignoring water vapor gives 31.2—a 3 % error that can mis-identify the gas.[DER]

### 3.1.3 Kinetic theory and molecular speeds

**[CORE]** Assumptions: many point-like molecules, random motion, elastic collisions, no forces except during collisions, Newtonian mechanics.

**FORMULA**

$$P = \tfrac{1}{3}\,n_N\,m\,\langle v^2\rangle, \qquad \langle \tfrac{1}{2}mv^2\rangle = \tfrac{3}{2}kT \tag{3.6}$$

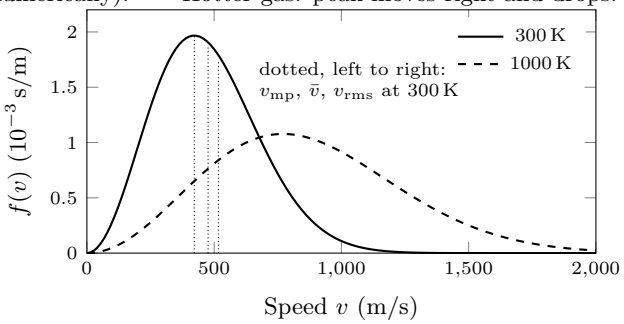
$$v_{\mathrm{rms}} = \sqrt{\frac{3RT}{M}}, \quad \bar{v} = \sqrt{\frac{8RT}{\pi M}}, \quad v_{\mathrm{mp}} = \sqrt{\frac{2RT}{M}} \tag{3.7}$$

$$f(v) = 4\pi\Big(\frac{m}{2\pi kT}\Big)^{3/2} v^2\, e^{-mv^2/2kT} \tag{3.8}$$

$n_N = N/V$ (number density, $\mathrm{m^{-3}}$), $m = M/N_A$ (kg per molecule). Ratios $v_{\mathrm{mp}} : \bar{v} : v_{\mathrm{rms}} = 1 : 1.128 : 1.225$. $M$ **in kg/mol** (0.028 kg/mol for $N_2$, not 28).

**Speed vs velocity.** Each velocity component is Gaussian with mean zero, $\langle v_x^2\rangle = kT/m$; the average *velocity* of a gas at rest is zero, while the average *speed* $\bar{v} > 0$. $f(v)$ is normalized: $\int_0^\infty f\,\mathrm{d}v = 1$; units s/m.

**Figure 3.1.** Maxwell speed distribution for $N_2$, computed from (3.8) (each curve has unit area, checked numerically).[DER] Hotter gas: peak moves right and drops.




**SOLVED DEMONSTRATION $N_2$ at 300 K, 1 atm.** $v_{\mathrm{mp}} = 422\,\mathrm{m/s}$, $\bar{v} = 476\,\mathrm{m/s}$, $v_{\mathrm{rms}} = 517\,\mathrm{m/s}$ (numerical moments of $f(v)$ agree). With collision diameter $d \approx 0.37\,\mathrm{nm}$ (approximate), $n_N = P/kT = 2.45\times10^{25}\,\mathrm{m^{-3}}$, mean free path $\lambda = 67\,\mathrm{nm}$, collision frequency $z = \bar{v}/\lambda = 7.1\times10^9\,\mathrm{s^{-1}}$.[DER]

### 3.1.4 Effusion, diffusion, mean free path

**FORMULA**

$$\Phi = \tfrac{1}{4}\,n_N\,\bar{v} \quad \text{(molecules per area per time)} \tag{3.9}$$

$$\frac{\mathrm{rate}_1}{\mathrm{rate}_2} = \sqrt{\frac{M_2}{M_1}} \quad \text{(Graham)} \tag{3.10}$$

$$\lambda = \frac{1}{\sqrt{2}\,\pi d^2 n_N} = \frac{kT}{\sqrt{2}\,\pi d^2 P} \tag{3.11}$$

**Effusion** = escape through a hole small compared with $\lambda$ (no collective flow). Graham's ratio requires the *same* $T$, same number density (pressure), and the same hole; the time to effuse equal amounts scales as $\sqrt{M}$. **Diffusion** = interpenetration through collisions; it depends on molecular size and the partner gas, so Graham's law is only a rough guide (hard-sphere estimate $D \sim \frac{1}{3}\lambda\bar{v}$).[EK]

**Example:** $H_2$ vs $O_2$ through the same pinhole at the same $T$ and $P$: rate ratio $\sqrt{32.00/2.016} = 3.98$; $O_2$ takes 3.98× as long to lose the same number of moles.[DER]

**Density trap:** $n_N$ in (3.11) is *number* density ($m^{-3}$), not molar density ($mol/m^3$); $n_N = N_A\,(n/V)$.

### 3.1.5 Equipartition and heat capacities

**[CORE]** Classically, each *active* quadratic energy term contributes $\frac{1}{2}kT$ per molecule ($\frac{1}{2}RT$ per mole). A fully active vibrational mode contributes two terms (kinetic + potential), i.e. $R$ per mole.

**FORMULA**

$$\bar{C}_p - \bar{C}_v = R \;\text{ (molar)}; \quad C_p - C_v = nR; \quad c_p - c_v = R_s \tag{3.12}$$

$$\gamma = \bar{C}_p/\bar{C}_v \tag{3.13}$$

**Table 3.2.** Rigid-molecule (vibration frozen) equipartition predictions.[EK]

| Molecule | dof (tr+rot) | $\bar{C}_v/R$ | $\bar{C}_p/R$ | $\gamma$ |
|---|---|---|---|---|
| Monatomic (He, Ar) | 3+0 | 1.5 | 2.5 | 5/3 |
| Linear ($N_2$, $O_2$, $CO_2$) | 3+2 | 2.5 | 3.5 | 7/5 |
| Nonlinear ($H_2O$, $CH_4$) | 3+3 | 3.0 | 4.0 | 4/3 |

Vibrational modes: $3N_a - 5$ (linear), $3N_a - 6$ (nonlinear) for $N_a$ atoms. **Freeze-out:** quantum level spacing larger than $kT$ makes a mode inactive. At 300 K, $N_2/O_2$ vibration is essentially frozen ($\bar{C}_p/R = 3.51, 3.54$), while $CO_2$ is *between* rigid (3.5) and fully excited (7.5) at 4.51 because its low-frequency bending modes are partly active (Tab. 3.3).[CP] So "$\gamma = 1.4$" is an idealization that drifts with temperature. Solids: 3 kinetic + 3 potential terms give $3R$ (Dulong–Petit).

**Table 3.3.** Gas properties at 300 K, 100 kPa from reference equations of state and transport models.[CP] *Steam row at 400 K (water is liquid at 300 K, 100 kPa).

| Gas | $M$ (g/mol) | $c_p$ (J/(kg · K)) | $\bar{C}_p/R$ | $\gamma$ | $k$ (mW/(m · K)) | $\mu$ (µPa · s) | $Z$ |
|---|---|---|---|---|---|---|---|
| He | 4.003 | 5193 | 2.500 | 1.667 | 156.0 | 19.93 | 1.0005 |
| Ar | 39.948 | 522 | 2.506 | 1.669 | 17.8 | 22.74 | 0.9994 |
| $H_2$ | 2.016 | 14 313 | 3.470 | 1.405 | 186.7 | 8.94 | 1.0006 |
| $N_2$ | 28.013 | 1041 | 3.509 | 1.401 | 26.0 | 17.89 | 0.9998 |
| $O_2$ | 31.999 | 920 | 3.540 | 1.396 | 26.5 | 20.65 | 0.9994 |
| Air (dry) | 28.965 | 1006 | 3.506 | 1.402 | 26.4 | 18.54 | 0.9997 |
| $CO_2$ | 44.010 | 853 | 4.513 | 1.293 | 16.8 | 15.00 | 0.9951 |
| $CH_4$ | 16.043 | 2236 | 4.314 | 1.305 | 34.4 | 11.25 | 0.9983 |
| $NH_3$ | 17.031 | 2162 | 4.429 | 1.315 | 25.1 | 10.16 | 0.9899 |
| $H_2O$ (400 K)* | 18.015 | 2008 | 4.350 | 1.331 | 26.8 | 13.28 | 0.9892 |

### 3.1.6 Real gases

**[CORE]** **Compressibility factor** $Z = P\bar{V}/(RT)$; $Z = 1$ ideal. $Z < 1$: attractions dominate (moderate $P$, lower $T$); $Z > 1$: excluded volume dominates (high $P$). Examples: $CO_2$ at 300 K: $Z = 0.995$ at 0.1 MPa, 0.687 at 5 MPa; $N_2$ at 300 K: 1.005 at 10 MPa, 1.364 at 50 MPa.[CP]

**FORMULA** van der Waals ($\bar{V} = V/n$):

$$\left(P + \frac{a}{\bar{V}^2}\right)(\bar{V} - b) = RT \tag{3.14}$$

$$\bar{V}_c = 3b, \quad T_c = \frac{8a}{27Rb}, \quad P_c = \frac{a}{27b^2}, \quad Z_c = \tfrac{3}{8} \tag{3.15}$$

$$a = \frac{27R^2T_c^2}{64P_c}, \qquad b = \frac{RT_c}{8P_c} \tag{3.16}$$

$a$: attraction (reduces wall pressure); $b$: excluded volume per mole. Units: $a$ in $Pa \cdot m^6/mol^2$ ($1 = 10\,L^2 \cdot bar/mol^2$); $b$ in $m^3/mol$ ($10^{-3}\,m^3/mol = 1\,L/mol$). Domain $\bar{V} > b$. Critical relations verified symbolically ($\partial P/\partial\bar{V} = \partial^2 P/\partial\bar{V}^2 = 0$).[DER] Real $Z_c$ values are 0.23–0.30 ($H_2O$ 0.229, $CO_2$ 0.275, $N_2$ 0.289),[CP] so vdW is qualitative near the critical point.

**Table 3.4.** van der Waals constants fitted to each gas's critical point.[CP][DER] Textbook tables fitted differently can disagree by a few percent.

| Gas | $T_c$ (K) | $P_c$ (MPa) | $a$ ($L^2 \cdot bar/mol^2$) | $b$ (L/mol) |
|---|---|---|---|---|
| He | 5.20 | 0.2283 | 0.0345 | 0.023 65 |
| $H_2$ | 33.14 | 1.2964 | 0.2471 | 0.026 57 |
| $N_2$ | 126.19 | 3.3958 | 1.3676 | 0.038 62 |
| $O_2$ | 154.60 | 5.0464 | 1.3813 | 0.031 84 |
| Ar | 150.69 | 4.8630 | 1.3618 | 0.032 20 |
| $CH_4$ | 190.56 | 4.5992 | 2.3028 | 0.043 06 |
| $CO_2$ | 304.13 | 7.3773 | 3.6565 | 0.042 85 |
| $NH_3$ | 405.56 | 11.3634 | 4.2214 | 0.037 09 |
| $H_2O$ | 647.10 | 22.0640 | 5.5348 | 0.030 48 |

**SOLVED DEMONSTRATION How good is van der Waals?** $CO_2$ at 300 K, 5.0 MPa. Ideal gas: $\bar{V} = RT/P = 0.499$ L/mol. vdW with Tab. 3.4 constants (cubic in $\bar{V}$; one real root here): $\bar{V} = 0.365$ L/mol, $Z = 0.731$. Reference equation of state: $Z = 0.687$.[CP] vdW captures most of the deviation (ideal: $Z = 1$) but not all; near the critical point it is only qualitative.[DER]

**Virial form** (extension): $Z = 1 + B(T)/\bar{V} + C(T)/\bar{V}^2 + \ldots$; vdW gives $B = b - a/(RT)$, so the **Boyle temperature** (where $B = 0$) is $T_B = a/(Rb)$.[DER] **[EXTENSION]**

**TRAP / LIMIT** An equation of state alone gives $P$–$\bar{V}$–$T$, not $c_p$, $h$, or $s$; caloric properties need ideal-gas heat capacities plus departure terms (or tables). For a vdW gas $U = nC_vT - an^2/V$, so even free expansion cools it slightly; an ideal gas does not cool.[DER]

## 3.2 First law: work, heat, processes, flow

### 3.2.1 Energy balance and sign conventions

**[CORE]**

**FORMULA** Closed system (this binder's default):

$$\Delta U = Q - W_{\text{by}} \qquad (\Delta E_{\text{sys}} = Q - W_{\text{by}} \text{ if KE, PE change}) \tag{3.17}$$

Chemistry form (IUPAC), $w$ = work done *on* the system:

$$\Delta U = q + w, \qquad w = W_{\text{on}} = -W_{\text{by}} \tag{3.18}$$

**Table 3.5.** One translation, used everywhere.

| Event | $Q$ | $W_{\text{by}}$ ($w = W_{\text{on}}$) |
|---|---|---|
| Heat enters system | + | |
| Gas expands, pushes surroundings | | + (−) |
| Gas compressed by surroundings | | − (+) |
| Stirring paddle, electric heater inside | | − (+) |

Cycle devices use **magnitudes** $Q_H, Q_L, W > 0$ with arrows (§4.1). A stirrer or resistor inside the system does work *on* it; that energy cannot be recovered as work (irreversible).

### 3.2.2 Boundary work

**[CORE]**

**FORMULA**

$$W_{\text{by}} = \int_{V_1}^{V_2} P_{\text{ext}}\,\mathrm{d}V \tag{3.19}$$

$$W_{\text{by}} = \int_{V_1}^{V_2} P\,\mathrm{d}V \quad (\text{quasistatic: } P_{\text{ext}} = P) \tag{3.20}$$

$$W_{\text{by}} = P_{\text{ext}}\,(V_2 - V_1) \quad (\text{constant } P_{\text{ext}}) \tag{3.21}$$

Use the gas's own $P$ only when the process is quasistatic (mechanical balance at every instant). A sudden expansion against a fixed external pressure uses $P_{\text{ext}}$; expansion into vacuum does *zero* work.

**Signed PV area.** On axes $P$ (up) vs $V$ (right): moving right gives $W_{\text{by}} > 0$ = area under the path; moving left gives $W_{\text{by}} < 0$. Around a cycle, $W_{\text{net,by}}$ = enclosed area, *positive for clockwise* traversal (engine), negative for counterclockwise (refrigerator/heat pump). On other axes ($V$ vs $P$, or $T$–$V$) the clockwise rule changes; recheck.

**FORMULA** Enthalpy and ideal-gas internal energy:

$$H \equiv U + PV; \quad Q_P = \Delta H \text{ (const. } P\text{, only } P\mathrm{d}V \text{ work)} \tag{3.22}$$

$$\Delta U = n\bar{C}_v\Delta T, \quad \Delta H = n\bar{C}_p\Delta T \quad \text{(ideal gas, } \textit{any} \text{ process)} \tag{3.23}$$

Ideal-gas $U$ and $H$ depend on $T$ only, so (3.23) holds even for non-constant-volume processes; constant $\bar{C}$ assumed (else integrate).

### 3.2.3 Process relations (ideal gas)

**[CORE]** The complete master table (work, heat, $\Delta U$, $\Delta S$, cues) is Tab. 1.2, p. 5.

**FORMULA**

$$\text{isothermal (rev.)}: W_{\text{by}} = Q = nRT\ln\frac{V_2}{V_1} = nRT\ln\frac{P_1}{P_2} \tag{3.24}$$

$$\text{adiabatic (rev.)}: PV^{\gamma},\ TV^{\gamma-1},\ TP^{(1-\gamma)/\gamma} \text{ const.} \tag{3.25}$$

$$W_{\text{by}} = \frac{P_1V_1 - P_2V_2}{\gamma - 1} = n\bar{C}_v(T_1 - T_2) \tag{3.26}$$

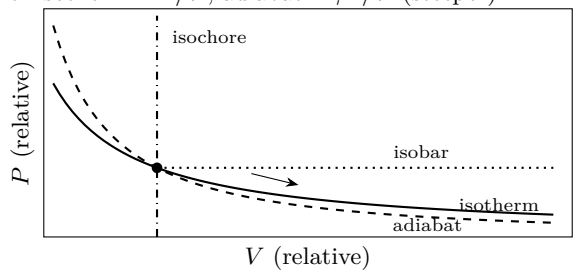
$$\text{polytropic } PV^{n_p}: W_{\text{by}} = \frac{P_2V_2 - P_1V_1}{1 - n_p} \quad (n_p \neq 1) \tag{3.27}$$

$$Q = n\bar{C}_n\Delta T, \quad \bar{C}_n = \bar{C}_v\,\frac{\gamma - n_p}{1 - n_p} \tag{3.28}$$

Polytropic special cases: $n_p = 0$ isobaric, 1 isothermal (use the logarithm), $\gamma$ adiabatic, $\infty$ isochoric. For $1 < n_p < \gamma$, $\bar{C}_n < 0$: the gas *warms while losing heat* (compressed faster than it can shed heat). Formula (3.28) verified for $n_p = 0, 1.2, 1.4, 3$.[DER]

**SOLVED DEMONSTRATION Adiabatic compression heats a gas.** Air ($\gamma = 1.4$) at 300 K, 100 kPa, compressed reversibly and adiabatically by a volume ratio of 18 (diesel-like): $T_2 = 300(18^{0.4}) = 953$ K (680 °C), $P_2 = 100(18^{1.4}) = 5.72$ MPa—hot enough to ignite injected fuel. *Check:* numerical integration of $\mathrm{d}T/T = -(\gamma - 1)\,\mathrm{d}V/V$ agrees. A bicycle pump warms for the same reason (only partly adiabatic).[DER]

**Figure 3.2.** Four reversible processes from a common state on $P$–$V$ axes (schematic, $\gamma = 1.4$). Slopes at the shared point: isotherm $-P/V$, adiabat $-\gamma P/V$ (steeper).[DER]




**SOLVED DEMONSTRATION Same endpoints, three paths.** Ideal gas from (200 kPa, 10 L) to (100 kPa, 20 L); both states at 300 K ($n = 0.8018$ mol), so $\Delta U = 0$ on every path and $Q = W_{\text{by}}$. (a) Reversible isotherm: $W = 2000 \ln 2 = 1386$ J. (b) Isochoric drop to 100 kPa, then isobaric: $W = 100(10) = 1000$ J. (c) Isobaric at 200 kPa, then isochoric drop: $W = 2000$ J. $Q$ and $W$ depend on path; $\Delta U = 0$ and $\Delta S = nR \ln 2 = 4.62$ J/K do not. *Check:* $\Delta S = Q_{\text{rev}}/T = 1386/300$ ✓.[DER]

**SOLVED DEMONSTRATION Reversible vs one-step compression.** 1.00 mol ideal gas at 300 K in contact with a 300 K reservoir, compressed 100 kPa→200 kPa (24.94 L→12.47 L). Reversible: $W_{\text{on}} = nRT \ln 2 = 1729$ J. One step with $P_{\text{ext}} = 200$ kPa throughout: $W_{\text{on}} = P_{\text{ext}}(V_1 - V_2) = 2494$ J. Same end state, so the extra 765 J leaves as extra heat to the reservoir: $S_{\text{gen}} = 2494/300 - R \ln 2 = 2.55$ J/K $> 0$. Irreversible compression always costs more work.[DER]

### 3.2.4 Steady-flow energy balance

**[CORE]** For one inlet (1) and one outlet (2) at steady state:

**FORMULA**

$$\dot{Q} - \dot{W}_{\text{sh}} = \dot{m}\left[h_2 - h_1 + \tfrac{V_2^2 - V_1^2}{2} + g(z_2 - z_1)\right] \quad (3.29)$$

$$\dot{m} = \rho A V \quad \text{(same at inlet and outlet)} \quad (3.30)$$

Flow work $Pv$ is already inside $h$. Units trap: $h$ in kJ/kg but $V^2/2$ in J/kg; 100 m/s is only 5 kJ/kg.

**Table 3.6.** Component selector (usual idealizations; drop terms only when justified).[EK]

| Device | Reduced balance |
|---|---|
| Heater, boiler, condenser | $\dot{Q} = \dot{m}(h_2 - h_1)$ ($\dot{W} = 0$) |
| Turbine (adiabatic) | $\dot{W}_{\text{out}} = \dot{m}(h_1 - h_2)$; $\eta_t = \dfrac{h_1 - h_2}{h_1 - h_{2s}}$ |
| Compressor (adiabatic) | $\dot{W}_{\text{in}} = \dot{m}(h_2 - h_1)$; $\eta_c = \dfrac{h_{2s} - h_1}{h_2 - h_1}$ |
| Pump (liquid) | $w_{\text{in}} \approx v\,\Delta P$ (incompressible, reversible) |
| Nozzle / diffuser | $\frac{1}{2}V_2^2 = \frac{1}{2}V_1^2 + h_1 - h_2$ |
| Throttling valve | $h_2 = h_1$ (adiabatic, no work, $\Delta$KE$\approx 0$) |
| Mixing chamber | $\sum \dot{m}_{\text{in}} h_{\text{in}} = \sum \dot{m}_{\text{out}} h_{\text{out}}$ |
| Heat exchanger | $\dot{m}_{\text{h}} \Delta h_{\text{h}} = \dot{m}_{\text{c}} \Delta h_{\text{c}}$ (insulated shell) |

Subscript $s$ = isentropic (ideal) outlet at the actual outlet pressure.

**SOLVED DEMONSTRATION Compressor with losses.** Air, $\dot{m} = 0.50$ kg/s, 300 K, pressure ratio 8, $\eta_c = 0.85$, $c_p = 1.005$ kJ/(kg · K), $\gamma = 1.4$. Ideal outlet $T_{2s} = 300(8^{0.2857}) = 543.4$ K; actual $T_2 = 300 + 243.4/0.85 = 586.4$ K. Power $\dot{W} = \dot{m} c_p (T_2 - T_1) = 143.9$ kW (ideal 122.3 kW). Irreversibility shows up as a *hotter* outlet and more work, never less.[DER]

### 3.2.5 Throttling, free expansion, Joule–Thomson

**[CORE]**

**FORMULA**

$$\mu_{\text{JT}} = \left(\frac{\partial T}{\partial P}\right)_h; \quad \mu_{\text{JT}} > 0 \Rightarrow \text{cools as pressure drops} \quad (3.31)$$

Ideal gas: $\mu_{\text{JT}} = 0$ (no temperature change on throttling). Real gases at 300 K, 0.1 MPa: $N_2$ +2.1, air +2.3, $CO_2$ +10.8, $H_2$ −0.30, He −0.63 K/MPa. Low-pressure inversion temperatures (sign change): $N_2$ 608 K, air 645 K, $H_2$ 201 K, He 46 K.[CP] So $H_2$ and He *warm* on throttling at room temperature and must be precooled below inversion before Joule–Thomson liquefaction. (vdW estimate $2a/(Rb) = 6.75\,T_c$ overestimates: 852 K for $N_2$.)

**Table 3.7.** Three "expansions" that are often confused.

| | Constraint | Ideal-gas result |
|---|---|---|
| Free expansion (into vacuum) | rigid insulated tank: $Q=0$, $W=0$, $\Delta U=0$ | $\Delta T=0$; $\Delta S = nR\ln\frac{V_2}{V_1}>0$ |
| Throttling (steady flow) | adiabatic valve: $h_2=h_1$ | $\Delta T=0$; $\Delta s = -R_s\ln\frac{P_2}{P_1}>0$ |
| Reversible adiabatic | work extracted, $\Delta S=0$ | $T_2 = T_1(P_2/P_1)^{(\gamma-1)/\gamma}$ (large drop) |

**TRAP / LIMIT** Not all cooling is Joule–Thomson. Vapor-compression refrigerators cool mainly because liquid refrigerant *flashes* (partly evaporates) across the valve into a low-pressure, low-$T_{\text{sat}}$ region; gas-cycle machines use expansion *turbines*; Peltier devices use the thermoelectric effect.

## 3.3 Second and third laws: entropy, free energy

### 3.3.1 Statements and reversibility

**[CORE]** **Kelvin–Planck:** no cyclic device can take heat from a single reservoir and convert it entirely to work. **Clausius:** no cyclic device can move heat from cold to hot with no other effect (no work input). A violation of either implies a violation of the other.

**Irreversibilities** (each generates entropy): heat transfer across a finite $\Delta T$, friction, unrestrained expansion, mixing of different substances, electrical resistance, inelastic deformation, chemical reaction away from equilibrium. A process is reversible only if both system *and* surroundings can be restored with no net change anywhere.

### 3.3.2 Entropy: definition and balance

**[CORE]**

**FORMULA**

$$\mathrm{d}S = \frac{\delta Q_{\text{rev}}}{T} \tag{3.32}$$

$$\oint \frac{\delta Q}{T_b} \le 0 \quad \text{(Clausius inequality)} \tag{3.33}$$

$$\Delta S_{\text{sys}} = \sum \int \frac{\delta Q}{T_b} + S_{\text{gen}}, \qquad S_{\text{gen}} \ge 0 \tag{3.34}$$

$T_b$ = absolute temperature *at the boundary where the heat crosses*. $S$ is a state function: to compute $\Delta S$ for an irreversible process, invent *any* reversible path between the same end states and integrate $\delta Q_{\text{rev}}/T$ along it. The actual $Q/T$ of the irreversible path is smaller; the difference is $S_{\text{gen}}$.

**Universe accounting:** $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = S_{\text{gen}}^{\text{total}} \ge 0$ (= 0 only if reversible). Over a full cycle $\Delta S_{\text{device}} = 0$, but $S_{\text{gen}} = -\sum_k Q_k/T_k \ge 0$ (heats into the device signed +) shows up in the reservoirs: **returning to the initial state does not make generation zero.**

**FORMULA** Entropy-change collection (state functions):

$$\Delta S = n\bar{C}_v \ln\frac{T_2}{T_1} + nR\ln\frac{V_2}{V_1} = n\bar{C}_p\ln\frac{T_2}{T_1} - nR\ln\frac{P_2}{P_1} \tag{3.35}$$

$$\Delta S = mc\ln\frac{T_2}{T_1} \quad \left(\text{general: } \int \frac{C(T)}{T}\mathrm{d}T\right) \tag{3.36}$$

$$\Delta S = \frac{\Delta H_{\text{tr}}}{T_{\text{tr}}} \quad \text{(phase change at equilibrium } T_{\text{tr}}\text{)} \tag{3.37}$$

$$\Delta S_{\text{res}} = \frac{Q_{\text{into res}}}{T_{\text{res}}} \quad \text{(reservoir: } T \text{ constant)} \tag{3.38}$$

$$\Delta S_{\text{mix}} = -nR\sum_i y_i \ln y_i \quad \text{(ideal gases, same } T, P\text{)} \tag{3.39}$$

Notes: (3.35) assumes constant $\bar{C}$; (3.37) only at the equilibrium transition temperature (e.g. melting ice at 0 °C: $333.4/273.15 = 1.221\,\text{J/(g·K)}$)—for supercooled water freezing at −10 °C, build a three-step reversible path (warm, freeze at 0 °C, cool). Mixing two samples of the *same* gas gives $\Delta S_{\text{mix}} = 0$ (Gibbs paradox resolved by indistinguishability).

**SOLVED DEMONSTRATION Entropy audit: hot copper into a lake.** 0.500 kg Cu ($c = 384\,\text{J/(kg·K)}$, $C = 192\,\text{J/K}$) at 100 °C into a lake at 15 °C (reservoir). Block: $\Delta S = 192\ln(288.15/373.15) = -49.6\,\text{J/K}$. Heat to lake: $Q = 192(85) = 16\,320\,\text{J}$, $\Delta S_{\text{lake}} = 16320/288.15 = 56.6\,\text{J/K}$. $S_{\text{gen}} = 7.0\,\text{J/K} > 0$ ✓. *Check:* numerical $\int C\,\mathrm{d}T/T$ matches; sign of $S_{\text{gen}}$ independent of which body is hotter.[DER]

**SOLVED DEMONSTRATION Supercooled water freezes at** −10 °C (per gram; take $c_w = 4.22$, $c_{\text{ice}} = 2.06\,\text{J/(g·K)}$ constant, $L_f = 333.4\,\text{J/g}$ at 0 °C). The real process is irreversible, so build a reversible path: warm liquid to 0 °C (+0.1574), freeze at 0 °C ($-333.4/273.15 = -1.2206$), cool ice to −10 °C (−0.0768): $\Delta S_{\text{sys}} = -1.1400\,\text{J/(g·K)}$. Heat released at −10 °C (Kirchhoff): $L = 333.4 - (4.22 - 2.06)(10) = 311.8\,\text{J/g}$; surroundings at 263.15 K: +1.1849. $S_{\text{gen}} = 0.045\,\text{J/(g·K)} > 0$: the system's entropy *falls*, yet the process is spontaneous.[DER]

**SOLVED DEMONSTRATION Mixing hot and cold water.** 1.00 kg at 80 °C + 1.00 kg at 20 °C, insulated, $c = 4.18\,\text{kJ/(kg}\cdot\text{K)}$: $T_f = 50\,^\circ\text{C}$. $\Delta S_{\text{hot}} = 4180\ln(323.15/353.15) = -371.1\,\text{J/K}$, $\Delta S_{\text{cold}} = 407.3\,\text{J/K}$, total 36.2 J/K generated by heat transfer across a finite $\Delta T$, even though no heat left the container.[DER]

### 3.3.3 T–S diagrams

For a *reversible* process, $Q = \int T\,\mathrm{d}S$ = area under the path on $T$–$S$ axes; a reversible cycle's enclosed area is net heat = net work (clockwise = engine with $T$ up, $S$ right). Reversible adiabatic = vertical line (isentropic); isothermal = horizontal line. For an irreversible path, the area is *not* the heat.

### 3.3.4 Statistical view

**FORMULA**

$$S = k\ln\Omega \tag{3.40}$$

$\Omega$ = number of microstates consistent with the macrostate. Doubling an ideal gas's volume multiplies $\Omega$ by $2^N$: $\Delta S = Nk\ln 2 = nR\ln 2$, matching (3.35). "Disorder" is a loose metaphor; entropy counts accessible microstates (how energy and matter can be distributed), and some ordered-looking states have higher entropy.[EK]

### 3.3.5 Free energies and spontaneity

**[CORE]**

**FORMULA**

$$A \equiv U - TS, \qquad G \equiv H - TS \tag{3.41}$$

$$\Delta G = \Delta H - T\Delta S \quad (\text{constant } T) \tag{3.42}$$

$$\mathrm{d}G \le 0 \quad (\text{const. } T, P;\ \text{no non-}P\mathrm{d}V\ \text{work}) \tag{3.43}$$

At constant $T, V$ use $\mathrm{d}A \le 0$. $-\Delta G$ = maximum non-expansion (e.g. electrical) work at constant $T, P$; $-\Delta A$ = maximum total work at constant $T$. "Spontaneous" means thermodynamically allowed, not fast.

**Table 3.8.** Sign logic for $\Delta G = \Delta H - T\Delta S$ (crossover $T^* = \Delta H/\Delta S$, if $\Delta H, \Delta S$ roughly constant).

| $\Delta H$ | $\Delta S$ | Favorable when |
|---|---|---|
| − | + | all $T$ |
| + | − | never (reverse is favorable) |
| − | − | low $T$ ($T < T^*$) |
| + | + | high $T$ ($T > T^*$) |

**FORMULA** Fundamental relations (closed, simple system):

$$\mathrm{d}U = T\mathrm{d}S - P\mathrm{d}V, \qquad \mathrm{d}H = T\mathrm{d}S + V\mathrm{d}P,$$
$$\mathrm{d}A = -S\mathrm{d}T - P\mathrm{d}V, \qquad \mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}P. \tag{3.44}$$

**[EXTENSION]** Maxwell relations: $(\partial T/\partial V)_S = -(\partial P/\partial S)_V$; $(\partial T/\partial P)_S = (\partial V/\partial S)_P$; $(\partial S/\partial V)_T = (\partial P/\partial T)_V$; $(\partial S/\partial P)_T = -(\partial V/\partial T)_P$.[DER]

### 3.3.6 Chemical potential and equilibrium

**FORMULA**

$$\mu_i = \left(\frac{\partial G}{\partial n_i}\right)_{T,P,n_{j\neq i}}, \quad \mu = \mu^\circ + RT\ln\frac{p}{p^\circ} \quad (\text{ideal gas}) \tag{3.45}$$

$$\Delta_r G = \Delta_r G^\circ + RT\ln Q \tag{3.46}$$

$$\Delta_r G^\circ = -RT\ln K \tag{3.47}$$

Matter moves from high to low $\mu$; coexisting phases have equal $\mu$ for each species. $Q$ and $K$ are built from **dimensionless activities**: gas $p_i/p^\circ$ ($p^\circ = 1\,\text{bar}$), solute $c_i/c^\circ$ ($c^\circ = 1\,\text{mol/L}$), pure solid or liquid = 1. $Q < K$: forward favorable; $Q = K$: equilibrium ($\Delta_r G = 0$). Temperature dependence: (6.6) p.29.

### 3.3.7 Third law

**[CORE]** As $T \to 0$, the entropy of a **perfect crystal with a unique (nondegenerate) ground state** approaches zero (Planck's form of Nernst's theorem). Consequences: absolute entropies

$$S^\circ(T) = \int_0^T \frac{C_p}{T}\mathrm{d}T + \sum \frac{\Delta H_{\text{tr}}}{T_{\text{tr}}} \tag{3.48}$$

are tabulated; $C_p \to 0$ as $T \to 0$; and **unattainability**: no finite sequence of processes reaches 0 K.

**Residual entropy:** glasses and orientationally disordered crystals keep $S > 0$ at $T \to 0$—e.g. CO (up to $R\ln 2$) and ice (Pauling estimate $R\ln\frac{3}{2} = 3.37\,\text{J/(mol}\cdot\text{K)}$).[EK][DER]

**TRAP / LIMIT** Elements in their reference state have $\Delta H_f^\circ = 0$ but $S^\circ > 0$ (graphite 5.7, $O_2$ 205.2 J/(mol · K)).[CRC] Never set an element's entropy to zero in $\Delta_r S^\circ$ sums.

**[EXTENSION]** **Landauer limit:** erasing one bit dissipates at least $kT\ln 2$ ($2.87 \times 10^{-21}$ J at 300 K). **Negative absolute temperature** occurs only in systems with bounded energy spectra (nuclear spins) with population inversion; such systems are "hotter" than any positive $T$. Not relevant to ordinary gases.

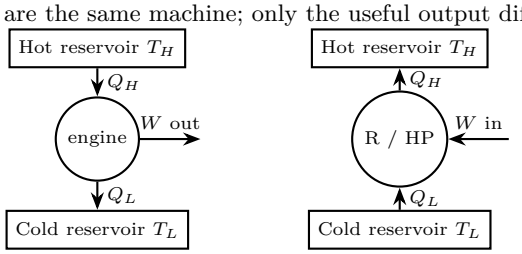
# TAB 4 Cycles: Engines, Refrigerators, Heat Pumps

## 4.1 Engines, refrigerators, heat pumps

### 4.1.1 Energy flows and performance

**[CORE]** All cycle quantities here are **positive magnitudes**; arrows carry the direction. Steady cyclic operation: $\Delta U_{\text{cycle}} = 0$, so $Q_H = W + Q_L$.

**Figure 4.1.** Energy-flow diagrams. A refrigerator and a heat pump are the same machine; only the useful output differs.




**FORMULA**

$$\eta = \frac{W}{Q_H} = 1 - \frac{Q_L}{Q_H} \tag{4.1}$$

$$\text{COP}_R = \frac{Q_L}{W}, \qquad \text{COP}_{HP} = \frac{Q_H}{W} = \text{COP}_R + 1 \tag{4.2}$$

$$\eta_C = 1 - \frac{T_L}{T_H} \tag{4.3}$$

$$\text{COP}_{R,C} = \frac{T_L}{T_H - T_L}, \qquad \text{COP}_{HP,C} = \frac{T_H}{T_H - T_L}$$

$\text{COP}_{HP} = \text{COP}_R + 1$ holds for the *same* machine with the same $Q_H, Q_L, W$ and all of $Q_H$ delivered as useful heat. Carnot bounds use the **reservoir** temperatures in kelvin—not working-fluid extremes of a different cycle, and never an average of endpoints. COP may exceed 1; $\eta$ may not.

**Second-law efficiency** $\eta_{II} = \eta/\eta_C$ (engines) or $\text{COP}/\text{COP}_C$ (refrigerators, heat pumps) measures how close a device comes to the reversible limit for the same reservoirs.

**SOLVED DEMONSTRATION Heat pump vs resistance heater.** Deliver 10.0 kW to a house. Resistance heater: 10.0 kW electric (COP = 1). Heat pump with $\text{COP}_{HP} = 3.0$: $W = 3.33$ kW, and $Q_L = 6.67$ kW comes from outdoors. Carnot limit for 35 °C supply and 0 °C outdoors: $308.15/35 = 8.8$, so $\eta_{II} = 0.34$.[DER]

**Finite (cooling) source.** A body of capacity $C$ cooling from $T_1$ to ambient $T_0$ can deliver at most $W_{\max} = C\big[(T_1 - T_0) - T_0 \ln(T_1/T_0)\big]$. For $C = 1$ kJ/K, $T_1 = 373.15$ K, $T_0 = 293.15$ K: 9.26 kJ of 80 kJ (11.6 %), not the 21.4 % a naive Carnot factor at $T_1$ suggests.[DER]

### 4.1.2 Machine-feasibility algorithm

**[CORE]**

**KEY IDEA / DECISION** (1) **Energy:** does $Q_H = W + Q_L$? If not, impossible (first law). (2) **Entropy:** $S_{\text{gen}} = \sum Q_{\text{out}}/T - \sum Q_{\text{in}}/T \geq 0$ over all reservoirs (engine: $Q_L/T_L - Q_H/T_H$; refrigerator: $Q_H/T_H - Q_L/T_L$). Negative: impossible; zero: reversible (Carnot); positive: possible but irreversible. Equivalent: $\eta \leq \eta_C$, $\text{COP} \leq \text{COP}_C$. (3) **Sanity:** kelvin temperatures, $T_H > T_L$, magnitudes positive, every reservoir counted.

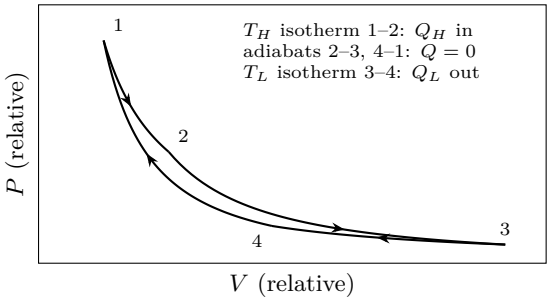
**SOLVED DEMONSTRATION Three claims**, reservoirs 500 K and 300 K, $Q_H = 1000$ kJ each. A: $W = 500$, $Q_L = 550$ kJ: $1000 \neq 1050$—violates energy. B: $W = 450$, $Q_L = 550$: energy OK; $S_{\text{gen}} = 550/300 - 1000/500 = -0.167$ kJ/K $< 0$ (equivalently $\eta = 0.45 > \eta_C = 0.40$): impossible. C: $W = 350$, $Q_L = 650$: $S_{\text{gen}} = 0.167$ kJ/K: possible, irreversible.[DER]

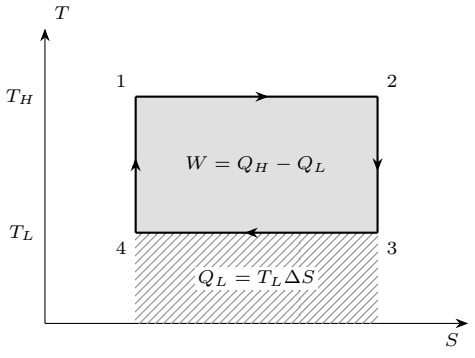
**Clockwise rule, stated carefully:** on $P$ (up)–$V$ (right) axes, or $T$ (up)–$S$ (right) axes, a clockwise loop has positive net work out (engine); counterclockwise = refrigerator/heat pump. Swapping axes reverses the rule.

### 4.1.3 Carnot cycle

**[CORE]** Reversible: isothermal expansion at $T_H$ (absorb $Q_H$) → adiabatic expansion to $T_L$ → isothermal compression at $T_L$ (reject $Q_L$) → adiabatic compression to $T_H$. For an ideal gas, $Q_H = nRT_H \ln(V_2/V_1)$, $Q_L = nRT_L \ln(V_3/V_4)$, and the adiabats force $V_2/V_1 = V_3/V_4$, giving $Q_L/Q_H = T_L/T_H$ and (4.3).

**Figure 4.2.** Carnot cycle. Left: $P$–$V$ computed for an ideal gas with $\gamma = 1.4$, $T_H/T_L = 5/3$, $V_2/V_1 = 2$ (relative units). Right: $T$–$S$ rectangle; enclosed area = $W$, area under the bottom edge = $Q_L$.[DER]



### 4.1.4 Air-standard piston and turbine cycles

**[CORE] Cold-air-standard assumptions:** working fluid is air as an ideal gas with constant $c_p$, $c_v$, $\gamma$; combustion replaced by heat addition, exhaust by heat rejection; closed, internally reversible processes. Real engines fall well short of these values. Ratios: **compression ratio** $r = V_{\max}/V_{\min}$; **cutoff ratio** $r_c = V_3/V_2$ (volume at end/start of constant-pressure heat addition); **pressure ratio** $r_p = P_{\text{high}}/P_{\text{low}}$.

**FORMULA**

$$\eta_{\text{Otto}} = 1 - \frac{1}{r^{\gamma-1}} \tag{4.4}$$

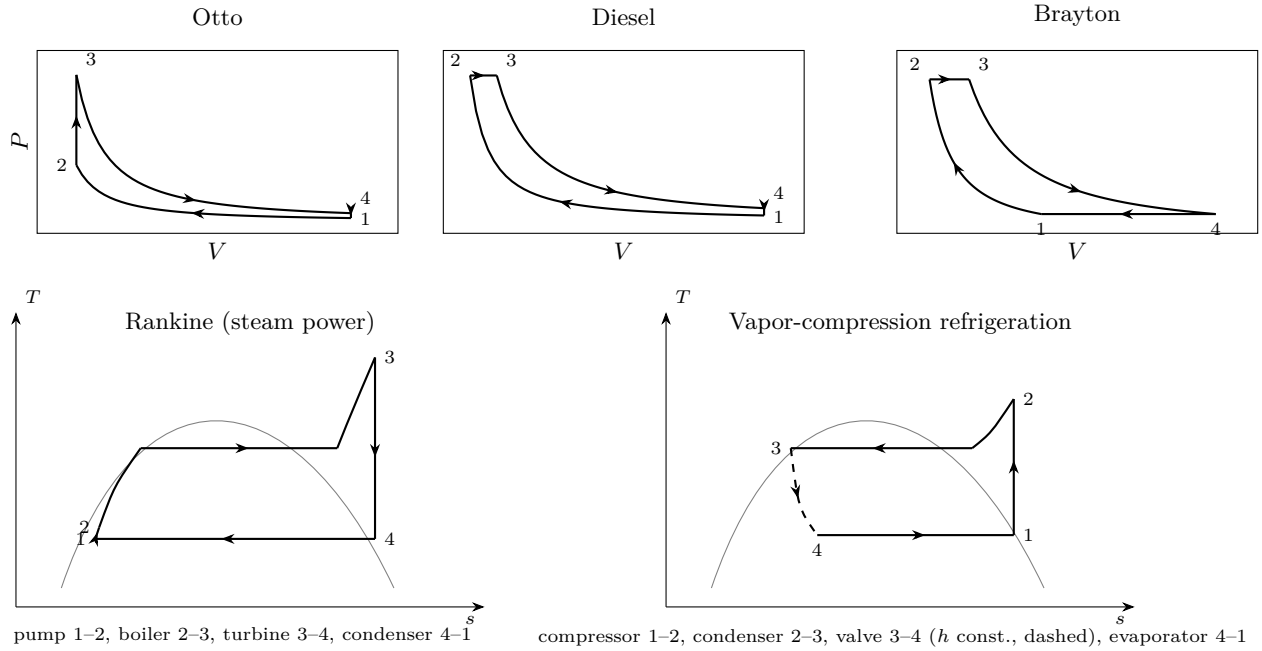
$$\eta_{\text{Diesel}} = 1 - \frac{1}{r^{\gamma-1}} \, \frac{r_c^{\gamma} - 1}{\gamma\,(r_c - 1)} \tag{4.5}$$

$$\eta_{\text{Brayton}} = 1 - \frac{1}{r_p^{(\gamma-1)/\gamma}} \tag{4.6}$$

All three verified against explicit state-by-state calculations.[DER] For the same $r$, Diesel < Otto (the bracket exceeds 1), but Diesels run at much higher $r$ ($r = 18$, $r_c = 2$: 63.2 %). Brayton ($r_p = 10$, $T_1 = 300\,\text{K}$, $T_3 = 1400\,\text{K}$): $\eta = 48.2\,\%$, back-work ratio $w_c/w_t = 0.41$—the compressor eats a large share of turbine work, so component efficiencies matter greatly. A regenerator (exhaust preheats compressed air) helps when $T_4 > T_2$.

**SOLVED DEMONSTRATION Otto vs Carnot between the same extremes.** $r = 8$, $\gamma = 1.4$, $T_1 = 300\,\text{K}$, $T_3 = 1800\,\text{K}$. $\eta = 1 - 8^{-0.4} = 56.5\,\%$; $T_2 = 300(8^{0.4}) = 689\,\text{K}$; $T_4 = 1800/8^{0.4} = 783\,\text{K}$. *Check:* $1 - (T_4 - T_1)/(T_3 - T_2)$ gives the same 56.5 %. A Carnot engine between 1800 K and 300 K would give 83.3 %: Otto adds heat over 689–1800 K and rejects it over 783–300 K.[DER]

**Figure 4.3.** Cycle shapes. Top: $P$–$V$ computed for $\gamma = 1.4$ (relative units; Otto $r = 8$; Diesel $r = 12$, $r_c = 2$; Brayton $r_p = 8$). Bottom: schematic $T$–$s$ with the vapor dome (not to scale). Numbers = state labels used in the text.[DER]




pump 1–2, boiler 2–3, turbine 3–4, condenser 4–1

compressor 1–2, condenser 2–3, valve 3–4 ($h$ const., dashed), evaporator 4–1

### 4.1.5 Vapor cycles: property-table method

**[CORE]** Phase change makes ideal-gas formulas useless; use $h$, $s$ from tables and the component balances of Tab. 3.6.

**FORMULA**

$$\eta_{\text{Rankine}} = \frac{(h_3 - h_4) - (h_2 - h_1)}{h_3 - h_2} \tag{4.7}$$

$$w_{\text{pump}} = h_2 - h_1 \approx v_1(P_2 - P_1)$$

$$\text{COP}_{R,\text{VCR}} = \frac{h_1 - h_4}{h_2 - h_1}, \qquad h_4 = h_3 \tag{4.8}$$

Tables: saturated water Tab. 2.7, superheated steam Tab. 7.8 (p. 36), R-134a Tab. 7.9 (p. 36). Isentropic turbine/compressor: find state 4 (or 2) from $s_4 = s_3$ (or $s_2 = s_1$) at the outlet pressure; inside the dome use $x = (s - s_f)/(s_g - s_f)$, then $h = h_f + x h_{fg}$.

**SOLVED DEMONSTRATION Ideal Rankine:** boiler 4 MPa, 400 °C; condenser 10 kPa. IAPWS-95 values (kJ/kg): $h_1 = 191.8$, $v_1 = 0.001\,010\,\text{m}^3/\text{kg}$, $w_p = v_1\Delta P = 4.03$, $h_2 = 195.8$; $h_3 = 3214.5$, $s_3 = 6.7714\,\text{kJ/(kg·K)}$; at 10 kPa $s_f = 0.6492$, $s_g = 8.1488$ $\Rightarrow x_4 = 0.816$, $h_4 = 2144.5$. $w_t = 1070.0$, $q_{\text{in}} = 3018.6$, $\eta = (1070.0 - 4.0)/3018.6 = 35.3\,\%$. Carnot between 673.15 K and $T_{\text{sat}}(10\,\text{kPa}) = 318.96\,\text{K}$: 52.6 %. *Check:* $q_{\text{in}} - w_{\text{net}} = h_4 - h_1 = 1952.7\,\text{kJ/kg}$ ✓; $h_4$ matches the library's $(P, s)$ flash.[IAPWS]

**SOLVED DEMONSTRATION Ideal vapor-compression, R-134a:** evaporator $-10\,^\circ$C (200.6 kPa), condenser $40\,^\circ$C (1016.6 kPa). $h_1 = 392.7$ (sat. vapor), $h_2 = 426.5$ ($s_2 = s_1$, $T_2 = 46.3\,^\circ$C), $h_3 = h_4 = 256.4$ (sat. liquid, then throttled; $x_4 = 0.338$). $\mathrm{COP}_R = (392.7 - 256.4)/(426.5 - 392.7) = 4.03$ vs Carnot $263.15/50 = 5.26$; as a heat pump $\mathrm{COP}_{HP} = 5.03 = \mathrm{COP}_R + 1$ ✓. Table reference states differ between sources: use only differences.[CP]

### 4.1.6 Multi-leg cycle ledger

**[CORE]** For any cycle drawn as legs, tabulate $Q$, $W_{\text{by}}$, $\Delta U$ per leg; checks: $\sum \Delta U = 0$, $\sum Q = \sum W_{\text{by}}$ = enclosed area; $\eta = W_{\text{net}}/\sum Q_{\text{in}}$ (only the *positive* $Q$ legs).

**SOLVED DEMONSTRATION Rectangle cycle**, monatomic ideal gas ($\Delta U = \frac{3}{2}\Delta(PV)$): 1 (100 kPa, 1 L) → 2 (300 kPa, 1 L) → 3 (300 kPa, 3 L) → 4 (100 kPa, 3 L) → 1.

| Leg | $\Delta U$ (J) | $W_{\text{by}}$ (J) | $Q$ (J) |
|---|---|---|---|
| 1–2 (const. $V$) | +300 | 0 | +300 |
| 2–3 (const. $P$) | +900 | +600 | +1500 |
| 3–4 (const. $V$) | −900 | 0 | −900 |
| 4–1 (const. $P$) | −300 | −200 | −500 |
| Sum | 0 | +400 | +400 |

$\eta = 400/1800 = 22.2\,\%$. Carnot between $T_3$ and $T_1$ ($T_3/T_1 = 9$): $88.9\,\%$.[DER]

### 4.1.7 Other cycles and effects

**Stirling** (two isotherms + two isochores) and **Ericsson** (two isotherms + two isobars) reach $\eta_C$ *only with ideal regeneration*: the heat released on one non-isothermal leg is stored and returned on the other, so external heat crosses only at $T_H$ and $T_L$. Without a perfect regenerator they fall below Carnot.

**[EXTENSION] Curzon–Ahlborn** $\eta_{CA} = 1 - \sqrt{T_L/T_H}$ is the efficiency at *maximum power* of one particular endoreversible model (finite-rate heat transfer through linear conductances). It is not a bound and not a universal law.

**Thermoelectric effects** (compact): *Seebeck*—a temperature difference across a junction of dissimilar conductors produces an emf (thermocouples, thermoelectric generators). *Peltier*—current through a junction absorbs or releases heat (solid-state coolers). *Thomson*—heat absorbed or released when current flows along a single conductor with a temperature gradient. Practical efficiencies are far below Carnot.[EK]

**Table 4.1.** Cycle comparison (ideal, cold-air-standard where applicable).

| Cycle | Process sequence | Ideal performance | Typical device / note |
|---|---|---|---|
| Carnot | isotherm $T_H$, adiabat, isotherm $T_L$, adiabat | $1 - T_L/T_H$ (4.3) p.20 | benchmark; impractical |
| Otto | $s$-compression, $V$-heat in, $s$-expansion, $V$-heat out | $1 - r^{1-\gamma}$ (4.4) p.21 | spark-ignition engine; $r = V_{\max}/V_{\min}$ |
| Diesel | $s$-compression, $P$-heat in, $s$-expansion, $V$-heat out | (4.5) p.21 | compression-ignition; $r_c = V_3/V_2$ |
| Brayton | $s$-compression, $P$-heat in, $s$-expansion, $P$-heat out | $1 - r_p^{-(\gamma-1)/\gamma}$ (4.6) p.21 | gas turbine, jet engine; $r_p = P_2/P_1$ |
| Rankine | pump, boiler ($P$), turbine ($s$), condenser ($P$) | tables (4.7) p.21 | steam power plant |
| Vapor compression | compressor ($s$), condenser ($P$), valve ($h$), evaporator ($P$) | tables (4.8) p.21 | refrigerator, A/C, heat pump |
| Stirling / Ericsson | 2 isotherms + 2 isochores / isobars | $\eta_C$ *with ideal regenerator* | external-combustion engines |

# TAB 5 Heat Transfer

## 5.1 Conduction and thermal networks

### 5.1.1 Fourier's law; flux vs rate

[CORE]

**FORMULA**

$$\mathbf{q}'' = -k\,\nabla T, \qquad q''_x = -k\frac{\mathrm{d}T}{\mathrm{d}x} \tag{5.1}$$

$$\dot{Q} = q''A = \frac{kA\,(T_1 - T_2)}{L} \quad \text{(plane wall, steady)} \tag{5.2}$$

$q''$ = heat **flux** (W/m$^2$); $\dot{Q}$ = heat **rate** (W); $k$ in W/(m · K). Heat flows down the temperature gradient (minus sign). Plane wall with constant $k$ and no generation: linear $T(x)$; a lower-$k$ layer carries a *steeper* gradient for the same flux.

### 5.1.2 Resistances and networks

[CORE] Steady, one-dimensional, no generation: $\dot{Q} = \Delta T/R_{\text{tot}}$.

**FORMULA**

$$R_{\text{wall}} = \frac{L}{kA}, \qquad R_{\text{conv}} = \frac{1}{hA}, \qquad R_{\text{rad}} = \frac{1}{h_r A} \tag{5.3}$$

$$R_{\text{cyl}} = \frac{\ln(r_2/r_1)}{2\pi k L} \tag{5.4}$$

$$R_{\text{sph}} = \frac{1}{4\pi k}\left(\frac{1}{r_1} - \frac{1}{r_2}\right) \tag{5.5}$$

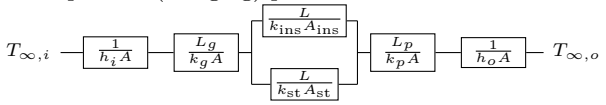
$$R_{\text{contact}} = R''_{tc}/A \tag{5.6}$$

$$\text{series: } R = \textstyle\sum R_i; \qquad \text{parallel: } \frac{1}{R} = \sum \frac{1}{R_i} \tag{5.7}$$

$$\dot{Q} = UA\,\Delta T_{\text{overall}}, \qquad UA = 1/R_{\text{tot}} \tag{5.8}$$

**Area basis of $U$.** For a cylinder, $U_iA_i = U_oA_o = 1/R_{\text{tot}}$ but $U_i \neq U_o$; always state which area. Convection and radiation from one surface act in *parallel* (same $\Delta T$ if surroundings ≈ fluid temperature).

**Figure 5.1.** Composite wall: films and layers in series; a stud forms a parallel (bridging) path. Each box is a resistance.




**Parallel-path assumption.** Treating stud and cavity as separate 1D paths assumes no sideways heat flow between them; the real 2D answer lies between this "parallel-path" result and an "isothermal-planes" estimate.[EK] **Thermal bridges** (studs, screws, metal frames, a beaker's support) are low-$R$ paths that short-circuit insulation.

**Contact resistance** arises from microscopic gaps at an interface; it falls with clamping pressure, soft interface materials, or thermal paste.

### 5.1.3 R-values (insulation ratings)

[CORE] Per-unit-area resistance $R'' = L/k$ (m$^2$ · K/W) is the SI "RSI". US $R$-value uses ft$^2$ · °F · h/Btu.

**FORMULA**

$$R_{\text{US}} = 5.678\,R_{\text{SI}}, \qquad R_{\text{SI}} = 0.1761\,R_{\text{US}} \tag{5.9}$$

(1 ft$^2$ · °F · h/Btu = 0.176 110 m$^2$ · K/W, IT Btu.)[PINT] R-values of layers in series add; $U'' = 1/R''_{\text{total}}$ including surface films. Parallel paths combine through $U$ (area-weighted), *not* by averaging $R$.

**SOLVED DEMONSTRATION Composite wall with studs.** Per m$^2$: inside film $h_i = 8$, outside $h_o = 25$ W/(m$^2$ · K) (given); gypsum 12.7 mm ($k = 0.16$); 89 mm cavity of mineral wool ($k = 0.040$) or pine stud ($k = 0.092$); plywood 12 mm ($k = 0.0907$). Resistances (m$^2$ · K/W): films 0.125 + 0.040; gypsum 0.079; plywood 0.132; wool 2.225; stud 0.967. Cavity path $R'' = 2.602$ (US R-14.8); stud path 1.344. With 15 % studs: $U'' = 0.85/2.602 + 0.15/1.344 = 0.438$ W/(m$^2$ · K), so whole-wall $R'' = 2.28$ (US R-13.0): bridging costs 12 %. At $\Delta T = 20$ K: $q'' = 8.8$ W/m$^2$. *Check:* conversion factor confirmed with a unit library; $R_{\text{stud}} < R_{\text{eff}} < R_{\text{cav}}$ ✓.[MAT][PINT]

### 5.1.4 Critical radius of insulation

Adding insulation to a *small* cylinder or sphere increases outer area (lower convection resistance) while adding conduction resistance. Heat loss is maximal at

$$r_{\text{cr,cyl}} = \frac{k_{\text{ins}}}{h}, \qquad r_{\text{cr,sph}} = \frac{2k_{\text{ins}}}{h}. \tag{5.10}$$

If the bare radius is below $r_{\text{cr}}$, thin insulation *increases* loss (e.g. $k = 0.04$, $h = 10$: $r_{\text{cr}} = 4$ mm—relevant for wires, not for a beaker).[DER]

**SOLVED DEMONSTRATION Where does an insulated container lose heat?** (Idealized geometry for illustration; not a design or rule-verified device.) Water at 60 °C, 0.400 kg, in a thin can of radius 4.0 cm, height 12 cm, wrapped radially in EPS ($k = 0.0355$ W/(m · K), Tab. 5.2) to radius 6.0 cm; 2.0 cm EPS discs top and bottom; still air at 22 °C; outer $\varepsilon = 0.90$; water-side film neglected. Side: $R_{\text{ins}} = \ln(6/4)/(2\pi \cdot 0.0355 \cdot 0.12) = 15.15$ K/W. Solving the outer-surface balance gives $T_o = 27.4$ °C, $h_c = 3.5$ (C6), $h_r = 5.4$ W/(m$^2$ · K), $R_{\text{out}} = 2.48$ K/W (insulation is 86 % of the side resistance). Each disc: $R \approx 122$ K/W. Total $K = 1/17.63 + 2/122 = 0.073$ W/K; $C = 1672$ J/K; $\tau = C/K = 6.3$ h; after 30 min: 57.1 °C. *Lessons:* with good insulation the outer finish barely matters; seams, lids, and supports (bridges) usually dominate real losses, and an open top adds evaporation (p. 12). Curvature check for C6 fails slightly here too, so $h_c$ is a mild underestimate.[DER][HT][CP]

### 5.1.5 Diffusivity, the heat equation, boundary conditions

**FORMULA**

$$\alpha = \frac{k}{\rho c_p} \quad (\mathrm{m^2/s}) \tag{5.11}$$

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} + \frac{\dot{q}}{\rho c_p} \quad (\text{1D, const. } k) \tag{5.12}$$

High $\alpha$ = temperature changes propagate quickly (metals); $k$ alone sets steady flux. Penetration depth after time $t$ scales as $\sqrt{\alpha t}$.

**Table 5.1.** Boundary conditions at a surface $x = 0$.[EK]

| Type | Condition |
|---|---|
| Specified temperature | $T(0,t) = T_s$ |
| Specified flux | $-k\,\partial T/\partial x\|_0 = q''_s$ |
| Insulated / symmetry | $\partial T/\partial x\|_0 = 0$ |
| Convection | $-k\,\partial T/\partial x\|_0 = h\,[T(0,t) - T_\infty]$ |
| Perfect-contact interface | $T$ and $q''$ continuous |

**[EXTENSION] Uniform generation** $\dot{q}$ (W/m$^3$), surfaces at $T_s$: plane wall of thickness $2L$: $T_{\max} = T_s + \dot{q}L^2/(2k)$; long cylinder of radius $r_0$: $T_{\max} = T_s + \dot{q}r_0^2/(4k)$. **Fins** help most where $h$ is small (gases) and fin $k$ is large; an infinitely long fin transfers $\sqrt{hPkA_c}\,(T_b - T_\infty)$.[EK]

**Table 5.2.** Thermal conductivity $k$, density $\rho$, and specific heat $c_p$ near room temperature. Building/insulation entries are *design values* for typical grades (not pure-element values); real products vary with density, moisture, and age.[MAT] Source codes: EN12524 = DIN EN 12524:2000; ASHRAE = Handbook Fundamentals 2013; air row from the Lemmon–Jacobsen model.[CP]

| Material | $k$ (W/(m · K)) | $\rho$ (kg/m$^3$) | $c_p$ (J/(kg · K)) | Source |
|---|---|---|---|---|
| Copper (design value) | 380 | 8900 | 380 | EN12524 |
| Aluminium alloys | 160 | 2800 | 880 | EN12524 |
| Steel (carbon) | 50 | 7800 | 450 | EN12524 |
| Stainless steel | 17 | 7900 | 460 | EN12524 |
| Cast iron | 50 | 7500 | 450 | EN12524 |
| Granite | 2.8 | 2600 | 1000 | EN12524 |
| Marble | 3.5 | 2800 | 1000 | EN12524 |
| Ice at 0 °C | 2.2 | 900 | 2000 | EN12524 |
| Quartz glass | 1.4 | 2200 | 750 | EN12524 |
| Soda-lime glass | 1 | 2500 | 750 | EN12524 |
| Brick, fired clay (1920 kg/m$^3$) | 0.895 | 1920 | 800 | ASHRAE |
| Water at 10 °C | 0.6 | 1000 | 4190 | EN12524 |
| PTFE | 0.25 | 2200 | 1000 | EN12524 |
| Polypropylene | 0.22 | 910 | 1800 | EN12524 |
| Acrylic (PMMA) | 0.2 | 1050 | 1500 | EN12524 |
| Oak | 0.17 | 705 | 1630 | ASHRAE |
| Polystyrene, solid | 0.16 | 1050 | 1300 | EN12524 |
| Gypsum board | 0.16 | 640 | 1880 | ASHRAE |
| Natural rubber | 0.13 | 910 | 1100 | EN12524 |
| Pine (oven-dried) | 0.092 | 370 | 1880 | ASHRAE |
| Cellular glass | 0.048 | 130 | 750 | ASHRAE |
| Mineral wool, 32 kg/m$^3$ | 0.04 | 32 | 840 | ASHRAE |
| Glass-fiber board | 0.036 | 160 | 840 | ASHRAE |
| EPS, molded beads | 0.0355 | 20 | 1470 | ASHRAE |
| XPS (extruded) | 0.026 | 32.5 | 1470 | ASHRAE |
| Polyisocyanurate, aged | 0.0235 | 30 | | ASHRAE |
| Still air, 300 K (gas property) | 0.0264 | 1.177 | 1006 | CoolProp |

## 5.2 Convection

### 5.2.1 Newton's cooling model and what sets $h$

**[CORE]**

**FORMULA**

$$q'' = h\,(T_s - T_\infty), \qquad \dot{Q} = hA_s\,(T_s - T_\infty) \tag{5.13}$$

$h$ (W/(m$^2$·K)) is **not a material property**: it depends on geometry and size, flow speed and regime, fluid properties, surface orientation, position (local $h_x$ vs average $\bar{h} = \frac{1}{L}\int_0^L h_x\,\mathrm{d}x$), and—for natural convection—on $\Delta T$ itself. Compute it from a correlation whenever geometry is known; avoid "typical $h$" lists.

**Natural (free)** convection: flow driven by buoyancy from density differences (governed by $Gr$, $Ra$). **Forced**: flow imposed by a fan, pump, or wind (governed by $Re$). Both matter when $Gr/Re^2 \sim 1$.[EK]

**Property evaluation.** External flows: at the *film temperature* $T_f = (T_s + T_\infty)/2$. Internal flows: at the bulk mean temperature (average of inlet and outlet mean temperatures). Ideal-gas $\beta = 1/T_f$ (kelvin). Air properties: Tab. 7.5; water: Tab. 7.6.

**Characteristic length.** Plate in parallel flow: length along the flow $L$. Cylinder or sphere in crossflow: diameter $D$. Vertical plate or tall cylinder (natural): height $L$. Duct: hydraulic diameter $D_h = 4A_c/P$. Horizontal plate (natural): use the convention of the chosen correlation's source (often $A_s/P$).[EK]

### 5.2.2 Dimensionless groups

**FORMULA**

$$Re = \frac{\rho V L}{\mu} = \frac{VL}{\nu}, \qquad Pr = \frac{\nu}{\alpha} = \frac{c_p \mu}{k} \tag{5.14}$$

$$Nu = \frac{hL}{k_{\text{fluid}}}, \qquad Bi = \frac{hL_c}{k_{\text{solid}}} \tag{5.15}$$

$$Gr = \frac{g\beta\,(T_s - T_\infty)L^3}{\nu^2}, \qquad Ra = Gr\,Pr = \frac{g\beta \Delta T L^3}{\nu\alpha} \tag{5.16}$$

Meanings: $Re$ inertia/viscous; $Pr$ momentum/thermal diffusivity ($\approx$ 0.7 gases, $\approx$ 7 water at 20 °C); $Nu$ convective enhancement over pure conduction in the fluid; $Gr$ buoyancy/viscous; $Bi$ internal conduction resistance/external convection resistance.

**TRAP / LIMIT** $Nu$ uses the **fluid's** $k$ and the correlation's length; $Bi$ uses the **solid's** $k$ and $L_c = V/A_s$ (lumped analysis, (5.26) p.27). They look alike and are not interchangeable.

**Regime markers (conventions, disturbance-dependent):** flat plate $Re_{x,c} \approx 5 \times 10^5$; pipe $Re_D \approx 2300$; vertical plate (natural) $Ra_L \approx 10^9$.[EK]

### 5.2.3 Choosing a correlation

**KEY IDEA / DECISION** (1) Natural or forced? (2) External or internal? (3) Geometry and orientation → length scale. (4) Boundary condition (isothermal vs uniform flux). (5) Compute $Re$ or $Ra$ with properties at the right temperature. (6) Check every validity limit (regime, $Pr$, $Re/Ra$ range, entrance length). (7) $h = Nu\,k/L$. (8) Add radiation in parallel when $h$ is small (natural convection in gases). Correlations: Tab. 5.3, p. 26.

**SOLVED DEMONSTRATION Beaker side wall: natural convection vs radiation vs wind.** Vertical cylinder $D = 7.0\,\text{cm}$, $H = 9.5\,\text{cm}$, $T_s = 60\,°\text{C}$, still air 22 °C. Film 314.15 K: $\nu = 1.710 \times 10^{-5}\,\text{m}^2/\text{s}$, $k = 0.027\,43\,\text{W/(m·K)}$, $Pr = 0.705$, $\beta = 1/314.15$. $Gr_H = 3.48 \times 10^6$, $Ra_H = 2.45 \times 10^6$ (laminar). C6: $Nu = 21.1$, $h_c = 21.1(0.02743)/0.095 = 6.1\,\text{W/(m}^2\cdot\text{K)}$. *Limit check:* needs $D \geq 35H/Gr^{1/4} = 7.7\,\text{cm}$; $D = 7.0\,\text{cm}$ fails slightly, so the flat-plate value somewhat *underestimates* $h_c$ (curvature thins the boundary layer). Radiation, $\varepsilon = 0.90$ (assumed), surroundings 22 °C: $h_r = 6.35\,\text{W/(m}^2\cdot\text{K)}$ ((5.21) p.27)—as large as convection. Side loss $\approx (6.1 + 6.35)(0.0209)(38) = 9.9\,\text{W}$, excluding top, bottom, and evaporation. Wind at 5 m/s (C3): $Re_D = 20\,500$, $Nu = 80$, $h = 31\,\text{W/(m}^2\cdot\text{K)}$, about 5× natural.[HT][CP][DER]

**TRAP / LIMIT** Do not apply an external-flow correlation inside a pipe, a laminar correlation in a turbulent range, or a "vertical plate" result to a short fat cylinder without the curvature check. Evaluate properties at the stated temperature. Natural-convection $h$ grows with $\Delta T$ (roughly $\Delta T^{1/4}$ in laminar flow), so "constant $h$" cooling models are approximations.

**Table 5.3.** Selected correlations. All $Nu$ are *averages* over the surface unless marked local. Forms cross-checked numerically against the `ht` library's implementations; validity notes from its documentation unless marked.[HT]

| Geometry, condition | Correlation | Validity; properties |
|---|---|---|
| C1 Flat plate, laminar, isothermal | $Nu_L = 0.664\,Re_L^{1/2}Pr^{1/3}$ (local: $Nu_x = 0.332\,Re_x^{1/2}Pr^{1/3}$) | $Re_L < 5\times10^5$; $0.6 < Pr < 10$; film $T$ |
| C2 Flat plate, turbulent from leading edge | $Nu_L = 0.037\,Re_L^{0.8}Pr^{1/3}$ (some texts: 0.036)[EK] | turbulent $Re_L$, $Pr \gtrsim 0.6$; film $T$ |
| C2′ Plate, laminar then turbulent | $Nu_L = (0.037\,Re_L^{0.8} - 871)\,Pr^{1/3}$ | $Re_{x,c} = 5\times10^5$ (871 derived from C1, C2)[DER] |
| C3 Cylinder in crossflow (Churchill–Bernstein) | $Nu_D = 0.3 + \dfrac{0.62\,Re_D^{1/2}Pr^{1/3}}{[1+(0.4/Pr)^{2/3}]^{1/4}}\left[1+\left(\dfrac{Re_D}{282\,000}\right)^{5/8}\right]^{4/5}$ | all $Re_D$ with $Re_DPr \gtrsim 0.2$ (tends to be a lower bound); film $T$ |
| C4 Pipe, laminar, fully developed | $Nu_D = 3.66$ (uniform $T_s$); $Nu_D = 4.36$ (uniform $q_s''$) | $Re_D < 2300$, beyond entrance length; bulk $T$[EK] |
| C5 Pipe, turbulent (Dittus–Boelter) | $Nu_D = 0.023\,Re_D^{0.8}Pr^n$, $n = 0.4$ heating fluid, 0.3 cooling | $Re_D \geq 10^4$, $0.6 \leq Pr \leq 160$, $L/D \geq 10$; bulk $T$ |
| C5′ Pipe, turbulent (Gnielinski) | $Nu_D = \dfrac{(f/8)(Re_D-1000)Pr}{1+12.7\sqrt{f/8}\,(Pr^{2/3}-1)}$, $f = (0.790\ln Re_D - 1.64)^{-2}$ | $2300 \leq Re_D \leq 5\times10^6$, $0.5 < Pr \leq 2000$; more accurate than C5 |
| C6 Vertical plate / tall cylinder, isothermal, natural (Churchill–Chu) | $Nu_L = \left\{0.825 + \dfrac{0.387\,Ra_L^{1/6}}{[1+(0.492/Pr)^{9/16}]^{8/27}}\right\}^2$ | all $Ra_L$ (laminar and turbulent); cylinder only if $D/L \geq 35/Gr_L^{1/4}$; film $T$ |
| C7 Horizontal cylinder, isothermal, natural (Churchill–Chu) | $Nu_D = \left\{0.60 + \dfrac{0.387\,Ra_D^{1/6}}{[1+(0.559/Pr)^{9/16}]^{8/27}}\right\}^2$ | $10^{-5} \leq Ra_D \leq 10^{12}$; film $T$ |
| C8 Horizontal plate, natural (McAdams) | hot face up / cold face down: $Nu = 0.54\,Ra^{1/4}$ ($Ra \leq 10^7$), $0.15\,Ra^{1/3}$ ($Ra > 10^7$); hot face down / cold face up: $Nu = 0.27\,Ra^{1/4}$ | length convention per source; film $T$ |
| C9 Sphere, isothermal, natural (Churchill) | $Nu_D = 2 + \dfrac{0.589\,Ra_D^{1/4}}{[1+(0.469/Pr)^{9/16}]^{4/9}} \times \Phi$ | $Ra_D < 10^{13}$; $\Phi = \{1 + 7.44\times10^{-8}Ra_D/[1+(0.469/Pr)^{9/16}]^{16/9}\}^{1/12} \approx 1$ for $Ra_D \lesssim 10^6$; $Nu \to 2$ as $Ra \to 0$ |

## 5.3 Thermal radiation

### 5.3.1 Emission and surface properties

**[CORE]**

**FORMULA**

$$E_b = \sigma T^4, \qquad E = \varepsilon\sigma T^4 \tag{5.17}$$

$$\alpha + \rho + \tau = 1 \quad \text{(opaque: } \tau = 0\text{)} \tag{5.18}$$

$$\lambda_{\max} T = b = 2897.77\,\mu\text{m}\cdot\text{K} \tag{5.19}$$

$\sigma = 5.670\,374\,419\times10^{-8}\ \text{W/(m}^2\cdot\text{K}^4\text{)}$ and $b$ follow exactly from the defined SI constants.[CODATA] $E_b$ = blackbody *emissive power* (emitted only, $\text{W/m}^2$); $G$ = irradiation (incident). $\varepsilon$ = emissivity, $\alpha$ = absorptivity, $\rho$ = reflectivity, $\tau$ = transmissivity. **Absolute temperature, fourth power**: doubling $T$ in kelvin multiplies emission by 16.

**Wien's law depends on the spectral variable.** (5.19) locates the peak of the *per-wavelength* distribution: Sun (5772 K) 0.50 µm; a 300 K surface 9.7 µm (infrared). The per-frequency distribution peaks at $\nu_{\max} = (5.879\times10^{10}\,\text{Hz/K})\,T$, which is *not* $c/\lambda_{\max}$.[CODATA][DER] Planck (per wavelength): $E_{b\lambda} = 2\pi hc^2/\{\lambda^5[\exp(hc/\lambda kT) - 1]\}$; its peak reproduces $b$ to 9 digits.[DER]

**Kirchhoff's law.** Spectral, directional $\varepsilon_{\lambda,\theta} = \alpha_{\lambda,\theta}$ holds for any surface. *Total* $\varepsilon = \alpha$ only if the irradiation comes from a blackbody at the surface's own temperature, or the surface is gray (properties independent of $\lambda$) and diffuse.

**TRAP / LIMIT Solar absorptivity is not infrared emissivity.** Sunlight peaks near 0.5 µm; room-temperature emission near 10 µm. A selective solar-collector coating has high $\alpha_{\text{solar}}$ and low $\varepsilon_{\text{IR}}$; white paint has low $\alpha_{\text{solar}}$ but high $\varepsilon_{\text{IR}}$. Visible color says little about IR emissivity. Glass transmits most sunlight but is nearly opaque to long-wave IR.[EK]

**Emissivity depends on surface condition** (finish, oxidation, coating, temperature, wavelength). Qualitative guide:[EK] polished metals (silver, aluminum foil, gold) are low (a few hundredths); oxidized or rough met-

als intermediate to high; most nonmetals—paints of any color, glass, water, skin, wood, paper—are high (about 0.85–0.95) in the IR. Use the value a problem gives.

### 5.3.2 Net exchange

**[CORE]**

**FORMULA** Small gray object (area $A$, $\varepsilon$) inside large surroundings at $T_{\text{sur}}$:

$$\dot{Q}_{\text{net}} = \varepsilon\sigma A\,(T_s^4 - T_{\text{sur}}^4) \tag{5.20}$$

Exact factorization as a coefficient (for networks):

$$h_r = \varepsilon\sigma\,(T_s + T_{\text{sur}})(T_s^2 + T_{\text{sur}}^2) \;\approx\; 4\varepsilon\sigma T_m^3 \tag{5.21}$$

The approximation (with $T_m$ = mean temperature) is good only when $T_s \approx T_{\text{sur}}$; for 60 °C vs 22 °C the exact and linearized $h_r$ differ by 0.4 %, but the gap grows quickly with $\Delta T$.[DER]

**View factor** $F_{ij}$ = fraction of diffuse radiation leaving surface $i$ that reaches $j$.

$$A_iF_{ij} = A_jF_{ji}\ \text{(reciprocity)}, \quad \textstyle\sum_j F_{ij} = 1\ \text{(enclosure)} \tag{5.22}$$

$F_{ii} = 0$ for flat or convex surfaces; a convex body completely enclosed by another has $F_{12} = 1$.

**FORMULA** Two diffuse-gray opaque surfaces forming an enclosure:

$$\dot{Q}_{12} = \frac{\sigma\,(T_1^4 - T_2^4)}{\dfrac{1-\varepsilon_1}{\varepsilon_1 A_1} + \dfrac{1}{A_1F_{12}} + \dfrac{1-\varepsilon_2}{\varepsilon_2 A_2}} \tag{5.23}$$

Large parallel plates ($A_1 = A_2$, $F_{12} = 1$):

$$q''_{12} = \frac{\sigma\,(T_1^4 - T_2^4)}{1/\varepsilon_1 + 1/\varepsilon_2 - 1} \tag{5.24}$$

Long concentric cylinders or spheres (1 inside 2):

$$\dot{Q}_{12} = \frac{\sigma A_1 (T_1^4 - T_2^4)}{\dfrac{1}{\varepsilon_1} + \dfrac{1-\varepsilon_2}{\varepsilon_2}\dfrac{A_1}{A_2}} \tag{5.25}$$

As $A_1/A_2 \to 0$, (5.25) reduces to (5.20): in a big room the walls' emissivity does not matter.

### 5.3.3 Radiation shields

A thin, low-$\varepsilon$ sheet between two surfaces adds two surface resistances and one space resistance. Parallel plates with *all* emissivities equal and $N$ shields: $q''_N = q''_0/(N+1)$. Low-$\varepsilon$ shields do far better than that.

**SOLVED DEMONSTRATION One shield.** Large plates at 400 K and 300 K, $\varepsilon = 0.80$: $q''_0 = \sigma(400^4 - 300^4)/(1/0.8 + 1/0.8 - 1) = 661.5\,\text{W/m}^2$. Insert a shield with $\varepsilon = 0.05$ on both faces: denominator $= (1/0.8 + 1/0.05 - 1) \times 2 = 40.5$, so $q''_1 = 24.5\,\text{W/m}^2$ (3.7 % of the original). *Check:* solving for the shield temperature (360.3 K) with equal fluxes on both sides gives the same $q''$.[DER]

**TRAP / LIMIT** Radiation is net exchange: a surface always emits $\varepsilon\sigma T^4$ even when it is colder than its surroundings. Use kelvin; $T^4$ in °C is meaningless. Emissive power and net exchange are different quantities. A shield works only if it is not in conductive contact with the surfaces it separates.

## 5.4 Transient heating and cooling

### 5.4.1 Lumped-capacitance model

**[CORE] Cue:** "cooling curve", "time to reach", "half-life", "time constant", "fitted slope".

**FORMULA** No input, constant $h$ and $T_\infty$, uniform body temperature:

$$\rho V c\,\frac{\mathrm{d}T}{\mathrm{d}t} = -hA_s\,(T - T_\infty)$$

$$\frac{T - T_\infty}{T_i - T_\infty} = e^{-t/\tau}, \qquad \tau = \frac{\rho V c}{hA_s} = \frac{C}{hA_s} = R_{\text{th}}C \tag{5.26}$$

$$t_{1/2} = \tau \ln 2, \qquad t = \tau \ln\frac{T_i - T_\infty}{T - T_\infty} \tag{5.27}$$

After $1\tau$: 36.8 % of the initial excess remains; $3\tau$: 5.0 %; $5\tau$: 0.7 %. $1/R_{\text{th}}$ may combine convection and radiation: $hA_s \to (h_c + h_r)A_s$, or any overall conductance $K$.

**Validity (small Biot number):**

$$Bi = \frac{hL_c}{k_{\text{solid}}} < 0.1, \qquad L_c = \frac{V}{A_s}. \tag{5.28}$$

$L_c$ conversions: plane wall of thickness $2L$ cooled on both faces: $L$; long cylinder radius $r_0$: $r_0/2$; sphere: $r_0/3$.

**TRAP / LIMIT** Heisler charts and one-term series solutions (distributed transients, an **[EXTENSION]** topic) define $Bi$ with the half-thickness $L$ or the radius $r_0$, *not* $V/A_s$. Convert before comparing: a sphere's $Bi_{r_0} = 3\,Bi_{V/A}$.

**SOLVED DEMONSTRATION Biot screening.** Copper sphere, $D = 1.0$ cm, $k = 380\,\text{W/(m}\cdot\text{K)}$, in air with $h = 10\,\text{W/(m}^2\cdot\text{K)}$: $L_c = r_0/3 = 1.67$ mm, $Bi = 4.4 \times 10^{-5} \ll 0.1$—lumped is excellent. For a cup of *water*, internal mixing by natural convection (not conduction) usually keeps the water nearly uniform; the lumped model then hinges on stirring, not on $k_{\text{water}}$.[DER]

### 5.4.2 Fitting a cooling curve

**[CORE]** Taking logs linearizes (5.26):

$$\ln\big(T - T_\infty\big) = \ln\big(T_i - T_\infty\big) - \frac{t}{\tau}. \tag{5.29}$$

Plot $\ln(T - T_\infty)$ vs $t$; slope $= -1/\tau$. Requirements and diagnostics: use the *measured* ambient (a wrong

$T_\infty$ bends the line); fit only the range you will predict in; inspect residuals; curvature signals $T$-dependent $h$ (natural convection, radiation), evaporation, or two time constants (container vs contents).

**SOLVED DEMONSTRATION ILLUSTRATIVE, NOT MEASURED.** Suppose $T_\infty = 22.0\,^\circ\text{C}$, $T(0) = 70.0\,^\circ\text{C}$, $T(10\,\text{min}) = 58.0\,^\circ\text{C}$. Then $\tau = 10/\ln(48/36) = 34.8\,\text{min}$, $t_{1/2} = 24.1\,\text{min}$, and $T(25\,\text{min}) = 22 + 48e^{-25/34.8} = 45.4\,^\circ\text{C}$. Two points give a fit but no uncertainty; real calibration needs repeated trials (§6.3).[DER]

### 5.4.3 Constant heat input

**[CORE]**

**FORMULA** $C\,\mathrm{d}T/\mathrm{d}t = P - K(T - T_a)$ with constant $P$, $K$, $C$, $T_a$:

$$T(t) = T_a + \frac{P}{K} + \left(T_0 - T_a - \frac{P}{K}\right)e^{-t/\tau}, \quad \tau = \frac{C}{K} \tag{5.30}$$

Limits: early slope $\mathrm{d}T/\mathrm{d}t \approx P/C$ (when $T \approx T_a$); steady rise $T_\infty - T_a = P/K$; $P = 0$ recovers (5.26). **Identifiability:** a cooling trace alone determines only $\tau = C/K$. Separating $C$ and $K$ needs a known $P$ (early slope gives $C$, plateau gives $K$) or an independently known $C$ (masses × specific heats).

**SOLVED DEMONSTRATION** $P = 50\,\text{W}$, $K = 0.50\,\text{W/K}$, $C = 600\,\text{J/K}$, start at $T_a = 22\,^\circ\text{C}$: $\tau = 1200\,\text{s}$; $T(600\,\text{s}) = 22 + 100(1 - e^{-0.5}) = 61.3\,^\circ\text{C}$ (numerical ODE agrees). The model's "steady state" $122\,^\circ\text{C}$ is unreachable for liquid water: boiling caps it. Check the physics before trusting an extrapolation.[DER]

**Sensor lag.** A probe with its own time constant $\tau_s$ obeys $\mathrm{d}T_s/\mathrm{d}t = (T - T_s)/\tau_s$; during a steady ramp at rate $a$ it reads low by $a\tau_s$. Wait several $\tau_s$ after insertion; small, well-coupled probes respond faster.

**[EXTENSION] Distributed transients** (large $Bi$): use one-term series or Heisler charts with their own $Bi$ and Fourier number $Fo = \alpha t/L^2$; not reproduced here.

**Table 5.4.** Applications: map each design feature to the mechanism and model term it changes.[EK]

| Feature | Mechanism affected | Model consequence |
|---|---|---|
| Vacuum gap (Dewar/vacuum flask) | removes gas conduction and convection | only radiation across the gap plus conduction through the neck and supports remain |
| Silvered or foil surfaces facing a gap | lowers $\varepsilon \Rightarrow$ radiation | (5.23)/(5.24) with small $\varepsilon$; foil pressed against a solid has no gap and acts only as a (good) conductor |
| Supports, neck, stopper, wires | conduction bridges | extra parallel $L/(kA)$ paths: make them long, thin, low-$k$ |
| Trapped air (foam, fiber, feathers) | still air $k \approx 0.026\,\text{W/(m·K)}$; small cells suppress convection | low effective $k$; compression or moisture raises it |
| Double glazing, gas gap | conduction across gas; convection if the gap is too wide; radiation | low-e coatings cut the radiation path |
| Lid or cover on hot water | stops evaporation and the rising plume | removes a latent loss that can exceed sensible losses (p. 12) |
| Fan, wind, stirring | raises $h$ (forced convection) | $h$ about 5× larger at 5 m/s in the beaker example |
| Dark/matte vs shiny exterior | $\alpha_{\text{solar}}$ and $\varepsilon_{\text{IR}}$ | collecting sunlight favors high $\alpha_{\text{solar}}$; retaining heat favors low $\varepsilon_{\text{IR}}$ |
| Added thermal mass | larger $C$ | longer $\tau = C/K$; slower change, but the same steady loss for the same $\Delta T$ |

# TAB 6 Thermochemistry, History & Device

## 6.1 Thermochemistry and chemical equilibrium

### 6.1.1 Conventions

**[CORE] Standard state** (°): pure substance in a specified phase at $p^\circ = 1\,\text{bar}$ (solutes: 1 mol/kg or 1 mol/L as stated). Temperature is *not* part of the definition; tables are usually at 298.15 K. **Formation reaction**: one mole of compound from its elements in their reference states (graphite, $O_2(g)$, $H_2(g)$, $N_2(g)$, $Br_2(l)$, ...), for which $\Delta H_f^\circ = \Delta G_f^\circ = 0$ but $S^\circ > 0$. Always write balanced equations **with phase labels**: $H_2O(l)$ and $H_2O(g)$ differ by 44.0 kJ/mol at 298.15 K.[CRC][IAPWS]

**FORMULA**

$$\Delta_r H^\circ = \sum \nu_p\, \Delta H_f^\circ(\text{prod}) - \sum \nu_r\, \Delta H_f^\circ(\text{reac}) \tag{6.1}$$

$$\Delta_r S^\circ = \sum \nu_p\, S^\circ(\text{prod}) - \sum \nu_r\, S^\circ(\text{reac}) \tag{6.2}$$

$$\Delta_r G^\circ = \sum \nu\, \Delta G_f^\circ = \Delta_r H^\circ - T\Delta_r S^\circ \tag{6.3}$$

Same pattern for $\Delta G_f^\circ$; for entropy use *absolute* $S^\circ$ (elements included).

**Hess's law:** $\Delta H$ is a state function, so reaction enthalpies add along any route. Reverse a reaction: change the sign. Multiply coefficients by $n$: multiply $\Delta H$ by $n$. Cancel species only if the phases match.

### 6.1.2 Calorimetry links

**FORMULA** Bomb ($q_V = \Delta U$) to enthalpy, ideal gases, condensed-phase $PV$ neglected:

$$\Delta_r H = \Delta_r U + \Delta n_{\text{gas}} RT \tag{6.4}$$

$\Delta n_{\text{gas}}$ = moles of gaseous products minus gaseous reactants per mole of reaction as written; condensed water counts zero.

**Heating values:** higher heating value (HHV) has product water liquid; lower (LHV) has water vapor; they differ by $n_{H_2O} \times 44.0\,\text{kJ/mol}$ at 25 °C.

**Bond-enthalpy estimate:** $\Delta_r H \approx \sum BE(\text{broken}) - \sum BE(\text{formed})$. Valid only for gas-phase species; average bond enthalpies carry errors of several kJ/mol per bond; misses resonance and phase changes. Prefer formation data when available; use the bond values a question supplies.

**Kirchhoff's law** (no phase change between $T_1$ and $T_2$):

$$\Delta_r H(T_2) = \Delta_r H(T_1) + \int_{T_1}^{T_2} \Delta_r C_p\, \mathrm{d}T. \tag{6.5}$$

**SOLVED DEMONSTRATION Methane combustion** at 298.15 K (Tab. 6.1 data). $CH_4(g) + 2\,O_2(g) \longrightarrow CO_2(g) + 2\,H_2O(l)$: $\Delta_r H^\circ = -393.5 + 2(-285.8) - (-74.6) = -890.5\,\text{kJ/mol}$ (HHV; 55.5 kJ/g). With $2\,H_2O(g)$: $-802.5\,\text{kJ/mol}$ (LHV); difference $88.0 = 2 \times 44.0$ ✓. Bomb: $\Delta n_{\text{gas}} = 1 - 3 = -2$, so $\Delta_r U = -890.5 + 2(8.314)(298.15)/1000 = -885.5\,\text{kJ/mol}$. $\Delta_r S^\circ = 213.8 + 2(70.0) - 186.3 - 2(205.2) = -242.9\,\text{J/K}$; $\Delta_r G^\circ$ from $\Delta G_f^\circ$ values: $-818.1\,\text{kJ/mol}$; from $\Delta H - T\Delta S$: $-818.1\,\text{kJ/mol}$ ✓.[CRC][DER]

### 6.1.3 Equilibrium and temperature

**[CORE]** $K = \exp(-\Delta_r G^\circ/RT)$ ((3.47) p.19); $\Delta_r G = \Delta_r G^\circ + RT \ln Q$ decides direction away from equilibrium.

**FORMULA** van 't Hoff (constant $\Delta_r H^\circ$ over the range):

$$\ln \frac{K_2}{K_1} = -\frac{\Delta_r H^\circ}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) \tag{6.6}$$

Exothermic ($\Delta_r H^\circ < 0$): $K$ falls as $T$ rises; endothermic: $K$ rises (Le Chatelier). A plot of $\ln K$ vs $1/T$ has slope $-\Delta_r H^\circ/R$.

**SOLVED DEMONSTRATION Ammonia synthesis**, $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$ at 298.15 K: $\Delta_r G^\circ = 2(-16.4) = -32.8\,\text{kJ/mol}$, so $K = e^{32800/(8.314\times298.15)} = 5.6 \times 10^5$. $\Delta_r H^\circ = -91.8\,\text{kJ/mol}$, $\Delta_r S^\circ = 2(192.8) - 191.6 - 3(130.7) = -198.1\,\text{J/K}$ (*check:* $\Delta H - T\Delta S = -32.7\,\text{kJ/mol}$, within table rounding). Both negative: favorable only below $T^* \approx 463\,\text{K}$; van 't Hoff estimate at 700 K: $K \approx 3 \times 10^{-4}$. Industry still runs hot—for *rate*—and compensates with high pressure (Le Chatelier).[CRC][DER]

**TRAP / LIMIT** Use consistent units ($\Delta G$ in J/mol with $R = 8.314$); $K$ is dimensionless (activities, p. 19); $\Delta_r G^\circ$ refers to standard states, not actual mixtures; a large $K$ says nothing about speed.

**Table 6.1.** Standard formation data at 298.15 K, 1 bar (CRC Handbook values as transcribed).[CRC] ‡Independent datasets differ by > 1 kJ/mol (ethanol(l) −277.0 to −276.0; NO 90.3–91.1; sucrose −2221.2).[XCHK] — = not in this dataset.

| Species | $\Delta H_f^\circ$ (kJ/mol) | $\Delta G_f^\circ$ (kJ/mol) | $S^\circ$ (J/(mol · K)) |
|---|---|---|---|
| $H_2O$(l) | −285.8 | −237.1 | 70.0 |
| $H_2O$(g) | −241.8 | −228.6 | 188.8 |
| $CO_2$(g) | −393.5 | −394.4 | 213.8 |
| CO(g) | −110.5 | −137.2 | 197.7 |
| $CH_4$(g) | −74.6 | −50.5 | 186.3 |
| $C_2H_6$(g) | −84.0 | −32.0 | 229.2 |
| $C_3H_8$(g) | −103.8 | −23.4 | 270.3 |
| $C_8H_{18}$(l) | −250.1 | — | — |
| $CH_3OH$(l) | −239.2 | −166.6 | 126.8 |
| $C_2H_5OH$(l) | −277.6‡ | −174.8 | 160.7 |
| $C_6H_6$(l) | 49.1 | 124.5 | 173.4 |
| $NH_3$(g) | −45.9 | −16.4 | 192.8 |
| NO(g) | 91.3‡ | 87.6 | 210.8 |
| $NO_2$(g) | 33.2 | 51.3 | 240.1 |
| $SO_2$(g) | −296.8 | −300.1 | 248.2 |
| HCl(g) | −92.3 | −95.3 | 186.9 |
| $CaCO_3$(s) | −1207.6 | −1129.1 | 91.7 |
| CaO(s) | −634.9 | −603.3 | 38.1 |
| NaCl(s) | −411.2 | −384.1 | 72.1 |
| $Fe_2O_3$(s) | −824.2 | −742.2 | 87.4 |
| $Al_2O_3$(s) | −1675.7 | −1582.3 | 50.9 |
| $C_{12}H_{22}O_{11}$(s) | −2226.1‡ | — | — |
| C (graphite) | 0.0 | 0.0 | 5.7 |
| $H_2$(g) | 0.0 | 0.0 | 130.7 |
| $O_2$(g) | 0.0 | 0.0 | 205.2 |
| $N_2$(g) | 0.0 | 0.0 | 191.6 |

## 6.2 History: people, experiments, ideas

**TRAP / LIMIT Verification status:** every entry below is editor knowledge.[EK] No primary or scholarly source could be accessed in this build (network-restricted), so dates are given at the precision the editor is confident of; "c." marks approximate dates. Confirm any date before relying on it to break a tie. Types: **E** experiment/observation, **P** publication/theory, **I** invention/patent, **N** later naming.

**Table 6.2.** Compact timeline (chronological).

| Year | Person(s) | Type | Contribution and connection |
|---|---|---|---|
| c. 1593 | Galileo Galilei (attribution debated) | I | Air thermoscope: shows hotter/colder without a scale; precursor of the thermometer. |
| 1714; 1724 | D. G. Fahrenheit | I, P | Mercury-in-glass thermometer; published his temperature scale (1724). |
| 1742 | Anders Celsius | P | Centigrade scale with *100 at the freezing point and 0 at boiling*; inverted to the modern orientation shortly after his death. |
| c. 1760s | Joseph Black | E | Latent heat and specific heat ("heat capacity"); distinguished quantity of heat from temperature. Lectures published posthumously (1803). |
| 1780s | A. Lavoisier & P.-S. Laplace | E | Ice calorimeter: measured heat by mass of ice melted. Lavoisier's caloric (heat-fluid) theory. |
| 1798 | Count Rumford (B. Thompson) | E | Cannon boring produced seemingly inexhaustible heat from friction; challenged caloric theory. |
| 1822 | Joseph Fourier | P | *Analytical Theory of Heat*: conduction law $q'' = -k\nabla T$ and the heat equation. |
| 1824 | Sadi Carnot | P | *Reflections on the Motive Power of Fire*: ideal reversible engine; efficiency limited by reservoir temperatures. |
| 1834 | Émile Clapeyron | P | Put Carnot's ideas in analytic form with the $P$–$V$ diagram; Clapeyron relation (later completed by Clausius). |
| 1842–1850 | J. R. Mayer; J. P. Joule; H. von Helmholtz | E, P | Mechanical equivalent of heat (Joule's paddle-wheel experiments, 1840s); conservation of energy (Mayer 1842, Helmholtz 1847). |
| 1848 | William Thomson (Lord Kelvin) | P | Absolute temperature scale independent of any substance. |
| 1850; 1865 | Rudolf Clausius | P, N | Clear statements of the first and second laws (1850); coined "entropy" and wrote $\mathrm{d}S = \delta Q_{\mathrm{rev}}/T$ (1865). |
| 1852 | Joule & Thomson | E | Porous-plug (throttling) experiments: Joule–Thomson effect. |
| 1859 | W. J. M. Rankine | P | *Manual of the Steam Engine*; the vapor power cycle now named for him. |
| 1860; 1872–77 | J. C. Maxwell; L. Boltzmann | P | Velocity distribution (Maxwell 1860); statistical entropy and the $H$-theorem (Boltzmann). The form $S = k \ln W$ and the symbol $k$ were written by Planck (c. 1900). |
| 1873 | J. D. van der Waals | P | Doctoral thesis: equation of state with attraction and excluded volume; continuity of liquid and gas. |
| 1875–1878 | J. W. Gibbs | P | *On the Equilibrium of Heterogeneous Substances*: chemical potential, phase rule, free energy. |
| 1876; 1890s | N. Otto; R. Diesel | I | Four-stroke spark-ignition engine (Otto); compression-ignition engine (Diesel patent, 1890s). |
| 1879; 1884 | J. Stefan; L. Boltzmann | E, P | Stefan's empirical $T^4$ law of radiation; Boltzmann's thermodynamic derivation. |
| 1893; 1900 | W. Wien; M. Planck | P | Displacement law; Planck's radiation law with energy quanta. |
| 1906; 1911 | W. Nernst; Planck | P | Nernst heat theorem → third law; Planck's $S \to 0$ form. |
| 1908 | H. Kamerlingh Onnes | E | First liquefaction of helium (using precooling plus Joule–Thomson expansion). |
| 2019 | CGPM (SI redefinition) | N | Kelvin defined by fixing $k = 1.380\,649 \times 10^{-23}$ J/K exactly.[CODATA] |

**Shared attribution:** energy conservation emerged from several workers (Mayer, Joule, Helmholtz, Colding); "Boyle's law" was also found by Mariotte; "Charles's law" was published by Gay-Lussac. Credit questions should be answered with the names the question's source uses.[EK]

## 6.3 Device: conditional modeling and calibration

**RULE-DEPENDENT Construction and procedure NOT verified.** The 2027 Rules Manual, official clarifications, and tournament instructions were not available to this build (official site blocked from the build environment; nothing attached). Unverified search *snippets* conflict: the official event description reportedly says teams build a device "designed to **collect** heat",[SO1][SO2] while a community wiki describes an insulated-beaker **heat-retention** task (hot water in a beaker, timed loss).[WIKI] Do not build, modify, or plan testing from this page. Read the official rules for: allowed materials and energy sources, dimensions, what is measured and when, scoring variables, required prediction or documentation, and impound. This section contains only physics that holds under either reading. Construction changes require rule review and fresh validation; this binder does not recommend an overnight rebuild.

### 6.3.1 Single-node energy balance

**[CORE]**

**FORMULA**

$$C_{\text{eff}}\frac{\mathrm{d}T}{\mathrm{d}t} = P_{\text{in}}(t) - K\,(T - T_{\text{amb}}) - \dot{Q}_{\text{other}} \qquad (6.7)$$

$C_{\text{eff}}$ = total heat capacity that moves with the measured temperature (water + inner container + probe; J/K); $K$ = overall loss conductance (W/K; $1/K$ = total network resistance, §5.1.2); $P_{\text{in}}$ = absorbed input power (zero for pure retention); $\dot{Q}_{\text{other}}$ = evaporation and other losses not proportional to $T - T_{\text{amb}}$.

**Assumptions:** one well-mixed temperature; constant $C_{\text{eff}}$, $K$, and $T_{\text{amb}}$ over the run (radiation and natural convection make $K$ rise slightly with $T$); no phase change; device parts not in the node either negligible or lumped into $K$.

**Solutions:** constant $P_{\text{in}}$: (5.30) p.28; no input: (5.26) p.27 with $\tau = C_{\text{eff}}/K$.

**KEY IDEA / DECISION What a trace can identify.** A cooling trace alone gives only $\tau = C_{\text{eff}}/K$. To get $K$ itself, compute $C_{\text{eff}}$ from masses × specific heats (Tab. 2.5, Tab. 7.2) or run a known-power heating test (early slope $= P/C_{\text{eff}}$, plateau rise $= P/K$).

### 6.3.2 Collection vs retention

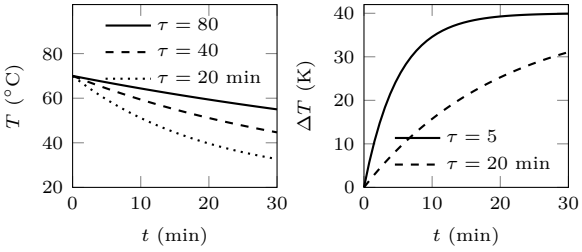
**Retention** (keep a hot load hot): minimize $K$ (every path in Tab. 5.4) and suppress evaporation. *Trap:* extra device mass that starts at room temperature initially *absorbs* heat from the load.

**Collection** (gain energy from a source): maximize absorbed $P_{\text{in}}$ (e.g. surface absorptivity for the source's spectrum, p. 26) and minimize $K$. With constant $P_{\text{in}}$ starting at ambient,

$$\Delta T(t) = \frac{P_{\text{in}}}{K}\left(1 - e^{-t/\tau}\right) \;\Rightarrow\; \begin{cases} \Delta T \approx P_{\text{in}}t/C_{\text{eff}}, & t \ll \tau \\ \Delta T \to P_{\text{in}}/K, & t \gg \tau \end{cases} \qquad (6.8)$$

So for a *short* window, small $C_{\text{eff}}$ gives the larger temperature rise; for a long window $K$ dominates. Energy stored $= C_{\text{eff}}\Delta T$ behaves differently from $\Delta T$. **Which quantity is scored is rule-dependent**—check before optimizing.[DER]

**Figure 6.1. ILLUSTRATIVE, NOT MEASURED**: symbolic model only. Left: retention from 70 °C in 22 °C ambient for three time constants. Right: collection with the same $P_{\text{in}}/K = 40$ K and two time constants $\tau = C_{\text{eff}}/K$.




### 6.3.3 Measurement, fitting, validation

**[CORE]** Universal good practice (not a rule):

1. **Preserve raw data**: time, temperature, ambient (start and end), water volume or mass, configuration, thermometer ID. Never overwrite readings with fitted values.
2. **Control** the measured quantity: same volume, probe position and depth, stirring before reading, lid state.
3. **Repeat** each condition (≥ 3 trials) and vary initial temperature across the range you may face; record ambient drift.
4. **Fit** a suitable model: $\ln(T - T_{\text{amb}})$ vs $t$ ((5.29) p.27), or an empirical calibration line of final temperature vs initial temperature at the specified duration (fewer assumptions, valid only inside the tested range).
5. **Check residuals** for curvature and outliers; check repeatability (standard deviation of repeats); correct for ambient by working with $T - T_{\text{amb}}$.
6. **Validate** on trials withheld from the fit; report prediction error.
7. **Report extrapolation separately**; note sensor lag (p. 28).

**Two different uncertainties.** For a straight-line fit with $n$ points, residual standard deviation $s$, the uncertainty of the *mean* response at $x_0$ scales as $s\sqrt{1/n + (x_0 - \bar{x})^2/S_{xx}}$, while a *single new trial* scatters by $s\sqrt{1 + 1/n + (x_0 - \bar{x})^2/S_{xx}}$ (a prediction interval, always wider; $S_{xx} = \sum(x_i - \bar{x})^2$).[EK] Predicting tomorrow's one run needs the second.

**TRAP / LIMIT Safety.** Hot water scalds; use tongs/mitts and stable bases. Never seal a container that will be heated (pressure build-up). No improvised electrical heaters or modified mains devices. Avoid thermal shock to glass (sudden large $\Delta T$). Follow every energy-source and material restriction in the official rules.

**Table 6.3.** Calibration record (blank template; one row per trial).

| Trial | Date/time | Setup / changes | Volume (mL) | $T_{\text{amb}}$ start/end (°C) | $T_0$ (°C) | $t$ (min) | $T_{\text{end}}$ (°C) | Prediction / notes |
|---|---|---|---|---|---|---|---|---|
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# TAB 7 Data, Error Prevention, Sources & Index

## 7.1 Data: constants, conversions, properties

**Table 7.1.** Where every data table lives.

| Data | Location |
|---|---|
| Water/ice/steam ledger constants | Tab. 2.5, p. 9 |
| Saturated water by $T$ / boiling point by $P$ | Tab. 2.7, p. 13; Tab. 2.8, p. 13 |
| Frost-point (ice) pressures | Tab. 2.9, p. 13 |
| Expansion coefficients | Tab. 2.3, p. 9 |
| Gases at 300 K ($c_p$, $\gamma$, $k$, $\mu$, $Z$) | Tab. 3.3, p. 15 |
| van der Waals constants | Tab. 3.4, p. 15 |
| Thermal conductivity of solids | Tab. 5.2, p. 24 |
| Formation enthalpy / Gibbs / entropy | Tab. 6.1, p. 30 |
| Constants, conversions, metals, air, liquid water | this section |
| Melting and boiling data, superheated steam, R-134a | this section |

**Table 7.2.** Specific heat of calorimetry metals at 298.15 K (CRC data as transcribed).[CRC] Last column tests Dulong–Petit ($\bar{C}_p/R \approx 3$).

| Metal | $M$ (g/mol) | $\bar{C}_p$ (J/(mol · K)) | $c_p$ (J/(g · K)) | $\bar{C}_p/R$ |
|---|---|---|---|---|
| Aluminum | 26.982 | 24.2 | 0.897 | 2.91 |
| Copper | 63.546 | 24.4 | 0.384 | 2.93 |
| Iron | 55.845 | 25.1 | 0.449 | 3.02 |
| Lead | 207.200 | 26.4 | 0.127 | 3.18 |
| Silver | 107.868 | 25.4 | 0.235 | 3.05 |
| Gold | 196.967 | 25.4 | 0.129 | 3.05 |
| Zinc | 65.380 | 25.4 | 0.388 | 3.05 |
| Nickel | 58.693 | 26.1 | 0.445 | 3.14 |
| Tin (white) | 118.701 | 25.8 | 0.217 | 3.10 |
| Titanium | 47.867 | 25.0 | 0.522 | 3.01 |
| Tungsten | 183.840 | 24.3 | 0.132 | 2.92 |

**Table 7.3.** Physical constants (CODATA 2022).[CODATA] *Exact by SI definition. †Exact because computed from exactly defined constants (shown rounded). Others: relative standard uncertainty $u_r$. $g_n$ and the standard atmosphere are conventional defined values.

| Constant | Value | Constant | Value |
|---|---|---|---|
| Boltzmann constant $k$ | $1.380\,649 \times 10^{-23}$ J/K* | Avogadro constant $N_A$ | $6.022\,140\,76 \times 10^{23}$ 1/mol* |
| Molar gas constant $R = N_A k$ | 8.314 462 618 J/(mol · K)† | Planck constant $h$ | $6.626\,070\,15 \times 10^{-34}$ J · s* |
| Speed of light $c$ | 299 792 458 m/s* | Stefan–Boltzmann $\sigma$ | $5.670\,374\,419 \times 10^{-8}$ W/(m$^2$ · K$^4$)† |
| Wien wavelength $b$ | $2.897\,771\,955 \times 10^{-3}$ m · K† | Wien frequency $b'$ | $5.878\,925\,758 \times 10^{10}$ Hz/K† |
| Elementary charge $e$ | $1.602\,176\,634 \times 10^{-19}$ C* | Faraday constant $F$ | $9.648\,533\,212 \times 10^{4}$ C/mol† |
| Standard gravity $g_n$ | 9.806 65 m/s$^2$* | Standard atmosphere | 101 325 Pa* |

**$R$ in common units:** 8.314 462 618 J/(mol · K) = 0.083 144 626 L · bar/(mol · K) = 0.082 057 37 L · atm/(mol · K) = 62.363 60 L · Torr/(mol · K) = 1.987 20 cal/(mol · K) (thermochemical cal) [CODATA][DER]

**Table 7.4.** Unit conversions computed with a unit library; the exact International Table Btu is defined explicitly.[PINT] *Exact by definition. Temperature *intervals*: 1 K = 1 °C = 1.8 °F = 1.8 R; scale conversions: (2.1) p.7.

| Quantity | Equals | Quantity | Equals |
|---|---|---|---|
| 1 thermochemical calorie | 4.184 J* | 1 bar | 100 kPa* |
| 1 International Table calorie | 4.1868 J* | 1 Torr | 133.3224 Pa |
| 1 Btu (International Table) | 1055.055 852 62 J* | 1 psi | 6.894 76 kPa |
| 1 kWh | 3.6 MJ* | 1 ft | 0.3048 m* |
| 1 eV | $1.602\,176\,634 \times 10^{-19}$ J* | 1 in | 2.54 cm* |
| 1 L·atm | 101.325 J* | 1 $ft^3$ | 28.3168 L |
| 1 hp (mechanical) | 745.699 87 W | 1 US gallon | 3.785 411 784 L* |
| 1 Btu/h | 0.293 07 W | 1 lbm | 0.453 592 37 kg* |
| 1 ton of refrigeration (12 000 Btu/h) | 3.5169 kW | 1 Btu/(lbm·°F) | 4186.8 J/(kg · K)* |
| 1 atm | 101.325 kPa* | 1 Btu/(h·ft·°F) | 1.7307 W/(m · K) |
| 1 atm | 760 Torr* | 1 Btu/(h·$ft^2$·°F) | 5.6783 W/($m^2$ · K) |
| 1 atm | 14.695 95 psi | 1 $ft^2$·°F·h/Btu (US R-1) | 0.176 11 $m^2$ · K/W |

**Table 7.5.** Dry air at 101.325 kPa.[CP] $\nu = \mu/\rho$, $\alpha = k/(\rho c_p)$; ideal-gas $\beta = 1/T$.

| $T$ (K) | $t$ (°C) | $\rho$ (kg/$m^3$) | $c_p$ (J/(kg·K)) | $k$ (mW/(m · K)) | $\mu$ (µPa · s) | $\nu$ ($mm^2$/s) | $\alpha$ ($mm^2$/s) | $Pr$ |
|---|---|---|---|---|---|---|---|---|
| 250 | −23 | 1.4133 | 1006 | 22.56 | 16.04 | 11.35 | 15.88 | 0.715 |
| 275 | 2 | 1.2843 | 1006 | 24.50 | 17.31 | 13.48 | 18.97 | 0.711 |
| 300 | 27 | 1.1770 | 1006 | 26.38 | 18.54 | 15.75 | 22.27 | 0.707 |
| 325 | 52 | 1.0863 | 1008 | 28.22 | 19.72 | 18.16 | 25.78 | 0.704 |
| 350 | 77 | 1.0085 | 1009 | 30.00 | 20.87 | 20.69 | 29.48 | 0.702 |
| 375 | 102 | 0.9412 | 1011 | 31.75 | 21.98 | 23.35 | 33.35 | 0.700 |
| 400 | 127 | 0.8823 | 1014 | 33.45 | 23.06 | 26.13 | 37.39 | 0.699 |
| 450 | 177 | 0.7842 | 1021 | 36.76 | 25.12 | 32.04 | 45.91 | 0.698 |
| 500 | 227 | 0.7057 | 1030 | 39.94 | 27.09 | 38.39 | 54.96 | 0.698 |

**Table 7.6.** Liquid water at 101.325 kPa.[IAPWS] $\beta$ = volumetric expansion coefficient (negative below 3.98 °C).

| $t$ (°C) | $\rho$ (kg/$m^3$) | $c_p$ (J/(kg · K)) | $k$ (W/(m · K)) | $\mu$ (mPa · s) | $\nu$ ($mm^2$/s) | $Pr$ | $\beta$ ($10^{-4}$ $K^{-1}$) |
|---|---|---|---|---|---|---|---|
| 5 | 999.97 | 4205 | 0.5678 | 1.5182 | 1.5182 | 11.24 | 0.16 |
| 10 | 999.70 | 4195 | 0.5788 | 1.3059 | 1.3063 | 9.47 | 0.88 |
| 20 | 998.21 | 4184 | 0.5980 | 1.0016 | 1.0034 | 7.01 | 2.07 |
| 25 | 997.05 | 4181 | 0.6065 | 0.8900 | 0.8927 | 6.14 | 2.57 |
| 30 | 995.65 | 4180 | 0.6144 | 0.7972 | 0.8007 | 5.42 | 3.03 |
| 40 | 992.22 | 4179 | 0.6285 | 0.6527 | 0.6578 | 4.34 | 3.85 |
| 50 | 988.04 | 4181 | 0.6406 | 0.5465 | 0.5531 | 3.57 | 4.58 |
| 60 | 983.20 | 4185 | 0.6510 | 0.4660 | 0.4740 | 3.00 | 5.23 |
| 70 | 977.76 | 4190 | 0.6598 | 0.4035 | 0.4127 | 2.56 | 5.84 |
| 80 | 971.79 | 4197 | 0.6670 | 0.3541 | 0.3643 | 2.23 | 6.41 |
| 90 | 965.31 | 4205 | 0.6728 | 0.3142 | 0.3255 | 1.96 | 6.97 |

**Table 7.7.** Melting points and enthalpies of fusion at about 1 atm (CRC data as transcribed; sorted by $T_m$).[CRC] The CRC water value (333.6 J/g) differs slightly from IAPWS (333.4 J/g, Tab. 2.5).

| Substance | $T_m$ (K) | $t_m$ (°C) | $\Delta\bar{H}_{\text{fus}}$ (kJ/mol) | $L_f$ (J/g) |
|---|---|---|---|---|
| Ethanol | 159.01 | −114.1 | 4.93 | 107.0 |
| Mercury | 234.32 | −38.8 | 2.29 | 11.4 |
| Water (ice) | 273.15 | 0.0 | 6.01 | 333.6 |
| Octadecane | 301.32 | 28.2 | 61.70 | 242.4 |
| Gallium | 302.91 | 29.8 | 5.58 | 80.1 |
| Tin | 505.08 | 231.9 | 7.15 | 60.2 |
| Lead | 600.61 | 327.5 | 4.77 | 23.0 |
| Zinc | 692.68 | 419.5 | 7.07 | 108.1 |
| Aluminum | 933.47 | 660.3 | 10.71 | 396.9 |
| NaCl | 1073.85 | 800.7 | 28.16 | 481.8 |
| Silver | 1234.93 | 961.8 | 11.30 | 104.8 |
| Gold | 1337.33 | 1064.2 | 12.55 | 63.7 |
| Copper | 1357.77 | 1084.6 | 13.26 | 208.7 |
| Iron | 1811.15 | 1538.0 | 13.81 | 247.3 |

**Table 7.8.** Superheated steam (IAPWS-95; same reference state as Tab. 2.7).[IAPWS]

| $t$ (°C) | $v$ ($m^3$/kg) | $h$ (kJ/kg) | $s$ (kJ/(kg · K)) |
|---|---|---|---|
| $P = 0.1$ *MPa*, $T_{sat} = 99.6$ °C | | | |
| 150 | 1.936 65 | 2776.6 | 7.6148 |
| 200 | 2.172 43 | 2875.5 | 7.8356 |
| 300 | 2.638 85 | 3074.5 | 8.2172 |
| 400 | 3.102 69 | 3278.6 | 8.5452 |
| 500 | 3.565 53 | 3488.7 | 8.8361 |
| 600 | 4.027 88 | 3705.6 | 9.0998 |
| $P = 1$ *MPa*, $T_{sat} = 179.9$ °C | | | |
| 200 | 0.206 02 | 2828.3 | 6.6955 |
| 300 | 0.257 99 | 3051.6 | 7.1246 |
| 400 | 0.306 61 | 3264.5 | 7.4669 |
| 500 | 0.354 11 | 3479.1 | 7.7641 |
| 600 | 0.401 11 | 3698.6 | 8.0310 |
| $P = 4$ *MPa*, $T_{sat} = 250.4$ °C | | | |
| 300 | 0.058 87 | 2961.7 | 6.3639 |
| 400 | 0.073 43 | 3214.5 | 6.7714 |
| 500 | 0.086 44 | 3446.0 | 7.0922 |
| 600 | 0.098 86 | 3674.9 | 7.3705 |

**Table 7.9.** Saturated R-134a (Tillner-Roth–Baehr equation of state). Reference state: $h = 200$ kJ/kg, $s = 1$ kJ/(kg · K) for saturated liquid at 0 °C; other tables may use other references—use differences only.[CP]

| $t$ (°C) | $P$ (kPa) | $h_f$ (kJ/kg) | $h_g$ (kJ/kg) | $s_f$ (kJ/(kg · K)) | $s_g$ (kJ/(kg · K)) |
|---|---|---|---|---|---|
| −30 | 84.4 | 160.8 | 380.3 | 0.8486 | 1.7515 |
| −20 | 132.7 | 173.6 | 386.6 | 0.9002 | 1.7413 |
| −10 | 200.6 | 186.7 | 392.7 | 0.9506 | 1.7334 |
| 0 | 292.8 | 200.0 | 398.6 | 1.0000 | 1.7271 |
| 10 | 414.6 | 213.6 | 404.3 | 1.0485 | 1.7221 |
| 20 | 571.7 | 227.5 | 409.7 | 1.0962 | 1.7180 |
| 30 | 770.2 | 241.7 | 414.8 | 1.1435 | 1.7145 |
| 40 | 1016.6 | 256.4 | 419.4 | 1.1905 | 1.7111 |
| 50 | 1317.9 | 271.6 | 423.4 | 1.2375 | 1.7072 |

**Table 7.10.** Normal boiling point (101.325 kPa), enthalpy of vaporization there, and critical point (reference equations of state).[CP]

| Fluid | $T_b$ (K) | $t_b$ (°C) | $h_{fg}$ (kJ/kg) | $\Delta\bar{H}_{\text{vap}}$ (kJ/mol) | $T_c$ (K) | $P_c$ (MPa) |
|---|---|---|---|---|---|---|
| He | 4.22 | −268.9 | 20.6 | 0.08 | 5.2 | 0.228 |
| $H_2$ | 20.37 | −252.8 | 448.7 | 0.90 | 33.1 | 1.296 |
| $N_2$ | 77.35 | −195.8 | 199.2 | 5.58 | 126.2 | 3.396 |
| Ar | 87.30 | −185.8 | 161.1 | 6.44 | 150.7 | 4.863 |
| $O_2$ | 90.19 | −183.0 | 213.1 | 6.82 | 154.6 | 5.046 |
| $CH_4$ | 111.67 | −161.5 | 510.8 | 8.20 | 190.6 | 4.599 |
| $C_3H_8$ | 231.04 | −42.1 | 425.6 | 18.77 | 369.9 | 4.251 |
| $NH_3$ | 239.83 | −33.3 | 1369.7 | 23.33 | 405.6 | 11.363 |
| R-134a | 247.08 | −26.1 | 217.0 | 22.14 | 374.2 | 4.059 |
| $C_4H_{10}$ | 272.66 | −0.5 | 385.7 | 22.42 | 425.1 | 3.796 |
| $C_3H_6O$ | 329.22 | 56.1 | 501.4 | 29.12 | 508.1 | 4.692 |
| $CH_3OH$ | 337.63 | 64.5 | 1101.1 | 35.28 | 513.4 | 8.216 |
| $C_2H_5OH$ | 351.57 | 78.4 | 849.6 | 39.14 | 514.7 | 6.268 |
| $C_6H_6$ | 353.22 | 80.1 | 393.7 | 30.75 | 562.0 | 4.906 |
| $H_2O$ | 373.12 | 100.0 | 2256.5 | 40.65 | 647.1 | 22.064 |

## 7.2 Error prevention and solution algorithms

**Table 7.11.** Error-prevention catalog: the symptom you see → the fix.

| Symptom / cue | Likely error | Fix or check | See |
|---|---|---|---|
| $T_f$ outside initial range (no input) | sign slip; missed phase change or input | signed ledger $\sum \Delta H_i = E_{\text{in}} - Q_{\text{loss}}$ | p. 9 |
| "$T_f > 0$ with ice left" | assumed phase state wrong | compare $Q_{\text{av}}$, $Q_1$, $Q_2$ first | p. 10 |
| Specific heat 5% low | container/probe capacity ignored | include $C_{\text{cal}}$ | p. 10 |
| Ratio, power, log, or $PV = nRT$ with °C | absolute temperature needed | convert to K (or R) | p. 7 |
| Pressure looks small / negative | gauge used as absolute | $P_{\text{abs}} = P_g + P_{\text{atm}}$ | p. 14 |
| $v_{\text{rms}}$ off by $\sqrt{1000}$ | $M$ in g/mol | $M$ in kg/mol | p. 14 |
| Mean free path off by $N_A$ | molar vs number density | $n_N = P/(kT)$ | p. 14 |
| Work with the wrong sign | mixed $W_{\text{by}}$ and $w = W_{\text{on}}$ | pick one convention, translate once | p. 16 |
| Work for sudden expansion | used gas $P$ instead of $P_{\text{ext}}$ | $W_{\text{by}} = \int P_{\text{ext}} \mathrm{d}V$; vacuum ⇒ 0 | p. 16 |
| $Q = n\bar{C}_p \Delta T$ at constant volume | $C_p/C_v$ swapped | constant $V$: $\bar{C}_v$; constant $P$: $\bar{C}_p$ | p. 16 |
| Adiabatic ⇒ "$T$ constant" | confused adiabatic and isothermal | adiabatic: $Q = 0$; use $PV^\gamma$ if reversible | p. 16 |
| Entropy of irreversible step via $Q/T$ | used actual heat | reversible path between the same states | p. 18 |
| $S_{\text{gen}} < 0$ | impossible process or sign error | recheck magnitudes and which reservoir gains | p. 18 |
| Carnot with °C or endpoint averages | wrong temperatures | reservoir temperatures in K | p. 20 |
| $\text{COP}_{HP} = \text{COP}_R + 1$ misapplied | different machines or lost heat | same machine, all $Q_H$ useful | p. 20 |
| "Clockwise = engine" on odd axes | axis orientation | rule holds for $P$–$V$ and $T$–$S$ only | p. 20 |
| Heat rate vs flux mismatch | W vs W/m$^2$ | multiply flux by the correct area | p. 23 |
| R-value conversion wrong by 32× | direction of 5.678 | $R_{\text{US}} = 5.678 R_{\text{SI}}$ | p. 23 |
| Parallel layers "averaged" | averaged $R$ | add conductances $UA$ | p. 23 |
| $Nu$ used as $Bi$ (or reverse) | fluid vs solid $k$ | $Nu$: fluid $k$; $Bi$: solid $k$, $L_c = V/A_s$ | p. 25 |
| Correlation outside range | regime/geometry mismatch | check validity column | p. 26 |
| $\sigma T^4$ with °C; "cold objects don't radiate" | absolute $T$; net exchange | kelvin; both surfaces emit | p. 27 |
| Solar $\alpha$ used as IR $\varepsilon$ | Kirchhoff misapplied | spectral match required | p. 26 |
| Cooling fit curves upward/downward | wrong $T_\infty$; nonconstant $h$ | measured ambient; residuals; fit range | p. 27 |
| Bomb $\Delta U$ reported as $\Delta H$ | constant-$V$ data | add $\Delta n_{\text{gas}} RT$ | p. 29 |
| Element entropy set to zero | confused $\Delta H_f^\circ$ with $S^\circ$ | elements: $\Delta H_f^\circ = 0$, $S^\circ > 0$ | p. 30 |
| $\ln K$ with units | non-dimensionless activities | $p/p^\circ$, $c/c^\circ$ | p. 19 |
| $L_v$(100 °C) for room-$T$ evaporation | latent heat varies with $T$ | use $h_{fg}(T)$ from table | p. 13 |

**KEY IDEA / DECISION A. Any energy problem.** (1) Draw the boundary; closed, open, or isolated? (2) List what crosses: $Q$, $W$ (type), mass. (3) Pick the balance: (3.17) p.16, (2.12) p.9, or (3.29) p.17. (4) Write state relations (ideal gas, tables, phase). (5) Solve symbolically, then substitute with units. (6) Check sign, range, limits, and units.

**KEY IDEA / DECISION B. Entropy audit.** (1) $\Delta S$ of each body via a reversible path (p. 18). (2) Reservoirs: $Q/T$. (3) Sum = $S_{\text{gen}} \geq 0$. (4) For machines, also check $\eta \leq \eta_C$ (p. 20).

**KEY IDEA / DECISION C. Heat-transfer network.** (1) Sketch nodes (temperatures) and paths. (2) Each path's resistance: (5.3) p.23–(5.6), convection from a correlation (Tab. 5.3), radiation via $h_r$ ((5.21) p.27). (3) Combine series/parallel. (4) $\dot{Q} = \Delta T/R$. (5) Back out intermediate temperatures; if $h$ or $h_r$ depended on a guessed surface $T$, iterate once.

**KEY IDEA / DECISION D. Transient.** (1) $Bi < 0.1$? ((5.28) p.27). (2) $C$ and $K$ (or $\tau$). (3) Input? Use (5.30) p.28; none: (5.26) p.27. (4) Sanity: physical limits (boiling, freezing), sensor lag.

## 7.3 Source register

**Access context (build of 2026-10-21):** the build environment's network policy blocked the Science Olympiad site, NIST, MIT, OpenStax, BIPM, and general web hosts (HTTP 403). No rules, host, or device documents were attached. Package registries were reachable, so authoritative *datasets and models* were used through versioned software and computed locally. A web-search tool returned result summaries (snippets) only; snippet-level items are labeled as such and are not treated as verified. Full provenance with exact functions and settings: `sources.md` and `data/provenance.json` in the source package.

**Table 7.12.** Source register (IDs appear as superscripts throughout).

| ID | Source | Access status | Used for |
|---|---|---|---|
| SO1 | Science Olympiad Inc., "Thermodynamics C" event page, `soinc.org/thermodynamics-c` | **not accessed** (403); search snippet only | conflict note on device purpose |
| SO2 | Science Olympiad Inc., 2027 Division C Event Slate (PDF) and 2027 conflict-block PDF, hosted under `soinc.org` (uploaded files) | **not accessed**; snippets only | event exists in 2027 slate (unverified) |
| WIKI | Scio.ly community guide "Thermodynamics", `scio.ly/docs/thermodynamics` | **not accessed**; snippet only; not authoritative | conflict note only |
| CODATA | CODATA 2022 recommended values, as distributed in SciPy 1.18.1 `scipy.constants` | inspected locally | constants, exact SI values |
| IAPWS | IAPWS-95 water formulation (Wagner & Pruß, *J. Phys. Chem. Ref. Data* 31, 387, 2002); viscosity (Huber et al. 2009), conductivity (Huber et al. 2012); via CoolProp 8.0.0 | computed locally; papers not accessed | water and steam properties, saturation tables |
| ICE | IAPWS-06 ice Ih and IAPWS (2011) melting/sublimation curves, via `iapws` 1.5.5 | computed locally; releases not accessed | ice $c_p$, $\rho$, $L_f$, frost-point pressures |
| CP | CoolProp 8.0.0 reference equations of state and transport models (e.g. $CO_2$ Span–Wagner 1996; air Lemmon et al. 2000 with Lemmon–Jacobsen 2004 transport; R-134a; $N_2$, $O_2$, Ar, He, $H_2$, $CH_4$, $NH_3$) | computed locally; papers not accessed | gas tables, critical points, $Z$, $\mu_{JT}$, R-134a states, air properties |
| CRC | CRC Handbook of Chemistry and Physics, standard thermodynamic properties table, as transcribed in `chemicals` 1.5.2 (`CRC_standard_data`); handbook edition not confirmed | package data inspected; handbook not accessed | $\Delta H_f^\circ$, $\Delta G_f^\circ$, $S^\circ$, metal $c_p$ |
| XCHK | Active Thermochemical Tables, NIST-JANAF, NIST WebBook values as transcribed in `chemicals` 1.5.2 | package data inspected | flags ($^\ddagger$) where datasets disagree |
| HT | `ht` 1.2.0 (C. Bell) correlation implementations and documentation, citing Churchill & Chu (1975, two papers), Churchill & Bernstein (1977), Dittus & Boelter (1930), Gnielinski (1976), McAdams, Bergman et al. (2011) | code and docstrings inspected; original papers not accessed | convection correlations and validity notes (formulas cross-checked numerically) |
| MAT | ASHRAE Handbook—Fundamentals (2013), DIN EN 12524:2000, VDI Heat Atlas (2010) design values, as transcribed in `ht` 1.2.0 `insulation` | package data inspected; one erroneous entry excluded | $k$, $\rho$, $c_p$ of solids and insulation |
| PINT | `pint` 0.26.2 unit definitions; exact IT Btu defined explicitly | inspected locally | unit conversions, R-value factor |
| DER | Derived or verified in this build: symbolic/numerical derivations; every solved demonstration recomputed with an independent check (`checks/demos.py`, 63 checks) | executed locally | equations marked DER and all demonstrations |
| EK | Editor knowledge: standard textbook results or facts *not* verified against an accessed source in this build | unverified this build | qualitative notes, some correlations, history, approximate coefficients |

**Recommended but not accessed (blocked):** official 2027 Rules Manual and clarifications (soinc.org); NIST Chemistry WebBook and CODATA web pages; J. H. Lienhard IV & V, *A Heat Transfer Textbook* (author-hosted); MIT OpenCourseWare thermodynamics notes; OpenStax *University Physics* Vol. 2. None of these is cited as

verification for any statement in this binder.

## 7.4 Index

Curated terms with common synonyms; numbers are clickable page links to substantive coverage. Also see the question router (p. 3) and formula selector (p. 4).

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